Journal of Logic and Computation, 2026, pp. 1–26
Paper

Recursive Completion in Higher λ-Models: Front-Seed Semantics, Proof-Relevant Witnesses, and the K_∞ Model

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Abstract

Martínez-Rivillas and de Queiroz gave extensional Kan semantics for the untyped λ-calculus and later constructed the concrete K_∞ homotopy λ-model. The two main mathematical results of the present paper are these. First, we show that a smaller front-seed coherence package—WLWR together with an inner-right-front pentagon contraction—already suffices to recover the associator comparison, semantic pentagon, and bridge theorems used in the later semantic arguments. Second, we prove explicit global reify, reflect, and application formulas for K_∞, with exact coordinatewise identities at every finite stage. We also record two structural clarifications: the recursive all-dimensional continuation of the explicit low-dimensional tower is obtained by a finite packaging phase followed by a uniform equality-generated recursion; and, on a deliberately fixed forward witness language for the classical β/η separation span, the canonical identity-type higher tower on K_∞ forces all higher non-connection once the two witness classes land at distinct points. The paper is fully formalized in Lean 4, and the project sources contain no local uses of `sorry`, `admit`, or `axiom`.

Keywords: higher lambda-calculus, proof relevance, extensional Kan complexes, globular coherence, denotational semantics, formal verification

2020 Mathematics Subject Classification. 03B40, 68Q55, 18N50.

1. Introduction

1.1. Background and lineage

The untyped λ-calculus has served since its creation as a meeting ground for logic, computation, and algebra. The classical denotational tradition, beginning with Scott's construction of continuous lattices [13] and codified in Barendregt's [2] systematic treatment, interprets λ-terms as elements of reflexive domains—ordered structures carrying reify and reflect maps. In that tradition the key equation $F(G(f))(x) = f(x)$ (and its dual) collapses the distinction between data and function, and the resulting models are *extensional* precisely when $G(F(x)) = x$. This interplay between the logical structure of convertibility and the computational content of reduction witnesses is a central theme of the present paper.

A broader motivation is the question whether conversion in a logical calculus can be organized into explicit higher witness data rather than only into a proposition-level equality. Identity types in homotopy type theory [6] and the computational-paths treatment of rewriting systems [4] provide familiar proof-relevant precedents, but we use them here only as orientation. The technical starting point is instead the work of Martínez-Rivillas and de Queiroz for the *untyped* calculus [9, 12, 11, 10], which interprets λ-terms in extensional Kan complexes, organizes higher $\beta\eta$-conversions into explicit n-cells, and recovers the classical conversion theory as the 0-truncation of that richer structure.

Against that background, the present paper asks a precise completion question: once Paper I provides the explicit low-dimensional higher-conversion core and Paper II provides the concrete K_∞ model together with its distinguished β/η witness pair, what all-dimensional structure is forced, what semantic coherence data are actually needed, and which parts of the K_∞ story can be made exact rather than merely existential? Two of the answers are substantive semantic/model-theoretic

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theorems (front-seed sufficiency and exact packaging), while the other two are structural clarifications about how the recursive completion and the canonical K_∞ tower behave once the baseline data are fixed.

1.2. The two baseline papers

We rely on two specific antecedents as a baseline; all results quoted from them are taken as established.

Paper I [11] introduced extensional Kan semantics for the untyped λ-calculus, defined the explicit tower of higher β- and η-conversions through dimension 3, and proved the central witness-lifting theorem: for every extensional Kan complex K, every explicit reduction witness lifts to a semantic 1-cell, and explicit syntactic 2- and 3-cells lift to semantic 2- and 3-cells between the corresponding interpreted witnesses. The proof proceeds by induction on the structure of reduction sequences, lifting individual-step β- and η-soundness to a full semantic interpretation of each dimension of the explicit tower. The resulting theory enriches the classical proposition-level equality $M =_{\beta\eta} N$ with a proof-relevant layer of semantic n-cells. Paper I also established $\beta\eta$-confluence via Hindley–Rosen and proved the invertibility of reduction sequences via Church–Rosser.

Paper II [10] constructed the K_∞ homotopy λ-model as the inverse limit $K_\infty = \varprojlim_n K_n$ of a tower of iterated function spaces over a flat base $K_0 = N^+$, where $N = \bigsqcup_{n\geq 0} S^n$. The construction follows the classical Scott tradition but enriches it with an explicit homotopical layer. Paper II proved that K_∞ is an algebraic bounded-complete domain (Proposition 4.1) and exhibited a separation phenomenon, recalled below in Example 8.1: the β-contraction and the η-contraction of a common redex land at the same target but receive semantically distinct images in K_∞. That example established the nontriviality of the proof-relevant structure but did not analyze the witness language at the level of explicit reduction terms or trace the separation through the recursively completed higher-cell tower. For provenance we continue to cite Paper II, but Sections 7 and 8 restate the specific ingredients used below: the projection-pair tower, the algebraicity/bounded-completeness proposition, and the fixed witness-separation span.

Table 1 makes the delta over the two baseline papers explicit.

Package	Inherited baseline	Gap filled here
Theorem A	Paper I supplies the explicit low-dimensional tower and recursive higher-derivation machinery.	We compare the explicit low-dimensional presentation with its all-dimensional continuation and prove strict globular boundary compatibility.
Theorem B	Associator/pentagon reasoning was available only through a stronger coherent semantic interface.	We isolate a smaller chosen-data interface sufficient for the later semantic comparison arguments.
Theorem C	Paper II isolated a distinguished β/η witness pair in K_∞.	We isolate a fixed forward witness language on that span and show that, in the canonical equality-generated tower on K_∞, the point-level separation persists automatically in every higher dimension.
Theorem D	Paper II constructed the inverse-limit model and its basic reflexive structure.	We prove exact global reify, reflect, and application formulas with two-sided equations.

Table 1 Exact delta over the inherited baseline.

1.3. Contributions of the present paper

Starting from those baseline results, this paper fills four specific gaps left by Papers I and II. The four theorem packages are not all of the same kind. Theorems 6.8 and 7.15 are the primary new generic-semantic and concrete-model results. Theorems 5.6 and 8.7 are structural clarifications: they explain what follows once the low-dimensional syntax and the point-level K_∞ separation are fixed.

Main contributions.

(A) **Comparison with the recursive completion (Section 5).** We record a structural comparison theorem for the explicit higher λ-conversion tower. Paper I supplies explicit data through dimension 3, and the recursive higher-derivation machinery already suggests how to continue above that range. The content of Theorem 5.6 is to compare the explicit low-dimensional presentation with the equality-generated continuation, identify the intermediate dimensions in which those presentations meet, and prove strict globular boundary compatibility for the resulting all-dimensional tower.

(B) **Front-seed sufficient semantic coherence (Section 6).** We isolate the precise refinement of the coherent semantic interface needed later: a bare extensional Kan complex equipped with a whiskering-left/whiskering-right comparison and only the inner-right-front pentagon contraction already suffices to recover the recursive associator comparison theorem, the semantic pentagon comparison for interpreted reduction sequences, and the explicit source bridge together with the target/shell bridges to the mixed target shell. The gain is that one specific smaller front-facing seed is sufficient for the later arguments while keeping WLWR explicit.

(C) **Fixed-span witness classification and separation (Section 8).** We refine the distinguished β/η witness pair already isolated in Paper II on the witness-separation span of Example 8.1 to a theorem about a deliberately fixed forward witness language on that span. The main new input is a canonical classification of those witnesses into β- and η-classes together with their canonical K_∞ interpretations. This is intentionally a fixed-span theorem rather than a global witness calculus: once the point-level dichotomy is fixed on that span, the higher-dimensional non-connection statements become structural consequences of the canonical equality-generated K_∞ tower.

(D) **Exact K_∞ reflexive packaging (Section 7.2).** We establish that the inverse limit K_∞ carries globally continuous reify, reflect, and application operations satisfying explicit stagewise formulas, yielding a complete reflexive c.h.p.o.. The novelty is not the abstract existence of a reflexive inverse limit, but the concrete formulas and exact coordinate identities needed later in the witness analysis.

The theorem labels are ordered by conceptual role, not by the narrative route through the paper. Theorems 6.8 and 7.15 carry the principal new semantic and model-theoretic input. Theorem 5.6 isolates the finite packaging boundary between the explicit low-dimensional syntax and the recursive continuation, while Theorem 8.7 records what the fixed-span point-level distinction implies inside the canonical higher tower on K_∞. Theorem 7.15 is developed before Theorem 8.7 because the witness-persistence argument in Section 8 uses the exact K_∞ packaging proved in Section 7.2.

1.4. Structure of the paper

Sections 2–4 collect definitions and restate the baseline results of Papers I and II. Sections 5–8 contain the new theorems, presented with detailed proof arguments. The narrative order is deliberate: Section 5 proves the structural tower comparison (Theorem 5.6), Section 6 isolates the smaller semantic interface (Theorem 6.8), Section 7.2 establishes the exact K_∞ model theorem (Theorem 7.15), and Section 8 returns to the fixed witness span to prove the final persistence theorem (Theorem 8.7). Section 9 gives a detailed comparison with related work and a list of open problems.

Sections 5 and 8 should be read as structural completion and persistence theorems: they explain what follows once the low-dimensional core and the point-level K_∞ separation are fixed. Sections 6 and 7.2 isolate the semantic interface and the model-specific inverse-limit structure on which those structural arguments rest.

For quick reference: Theorem 5.6 concerns only the syntactic tower, Theorem 6.8 concerns the generic semantic interface, Theorem 7.15 concerns the concrete inverse-limit model, and Theorem 8.7 combines the fixed-span witness tags with Theorem 7.15 inside the canonical K_∞ tower.

The mathematical content comes first throughout. The formal development [7] plays an evidential role: a brief verification note at the end of the paper records that all stated results have been mechanically checked, but the paper is self-contained and can be read without reference to the proof assistant. The constructive character of the proofs—particularly the explicit witness constructions in Theorems 5.6 and 8.7—reflects the paper's orientation toward the logic-and-computation tradition, where the computational content of proofs is a first-class object of study.

2. Preliminaries and Setup

2.1. Lambda-terms and reduction

We work with the untyped λ-calculus using de Bruijn indices. A *term* is generated by:

$$M, N ::= \mathtt{var}\ n \mid M\ N \mid \lambda\, M$$

where $n \in \mathbb{N}$ is a de Bruijn index. We write $M\ N$ for application and $\lambda\, M$ for abstraction. Substitution $M[N]$ denotes the capture-free replacement of the variable at index 0 in M by N, with the standard shifting conventions.

A *β-step* is a single reduction of the form $(\lambda\, M)\, N \to_\beta M[N]$, closed under subterm contexts. An *η-step* is a single reduction of the de Bruijn form

$$\lambda\, ((\mathsf{shift}(1,0)M)\, (\mathtt{var}\ 0)) \to_\eta M,$$

equivalently

$$\lambda\, (M\, (\mathtt{var}\ 0)) \to_\eta \mathsf{shift}(-1,0)M,$$

when $\mathtt{var}\ 0$ does not occur free in M. A *$\beta\eta$-step* is either a β-step or an η-step.

Definition 2.1 (Reduction sequences) A *reduction sequence* from M to N is a finite sequence of $\beta\eta$-steps and their formal inverses:

$$M = M_0 \leftrightarrow M_1 \leftrightarrow \cdots \leftrightarrow M_k = N.$$

We write $\mathrm{RedSeq}(M, N)$ for the type of all such sequences. Each reduction sequence is an explicit proof-relevant witness for the $\beta\eta$-convertibility of M and N.

2.2. Extensional Kan complexes

The semantic setting is that of *Kan complexes* in the simplicial sense.

Definition 2.2 (Kan complex) A *Kan complex* K is a simplicial set equipped with a chosen horn filler: for every horn $\Lambda \subset \Delta^{n+1}$ with a missing face, a chosen $(n+1)$-simplex σ extending the given faces. Formally, K carries a function

$$\mathsf{fill} : \mathrm{Horn}(K, n, j) \to K_{n+1}$$

satisfying the face-matching specification for all non-missing faces.

Definition 2.3 (Reflexive and extensional Kan complexes) A Kan complex K is *reflexive* if it is equipped with maps $F : |K| \to [K \to K]$ and $G : [K \to K] \to |K|$ satisfying the η-law $F(G(f))(x) = f(x)$. It is *extensional* if additionally $G(F(x)) = x$ for all objects x. Here $[K \to K]$ denotes the endofunction space on 0-simplices used by the semantic interpretation, and $|K|$ is the set of 0-simplices.

An extensional Kan complex K interprets λ-terms via:

$$[\![\mathtt{var}\ n]\!]_\rho = \rho(n), \qquad [\![M\ N]\!]_\rho = F([\![M]\!]_\rho)([\![N]\!]_\rho), \qquad [\![\lambda\, M]\!]_\rho = G\big(x \mapsto [\![M]\!]_{\rho[0\mapsto x]}\big).$$

Definition 2.4 (The theories $\mathrm{TH}_{\lambda=}$, $\mathrm{TheoryEq}_K$, and the proof-relevant tower over K) Fix an extensional Kan complex K. The *model-specific equality theory* is

$$\mathrm{TheoryEq}_K(M, N) \ :\equiv\ \forall\, \rho,\ [\![M]\!]_\rho = [\![N]\!]_\rho.$$

Separately, the *classical λ-theory* $\mathrm{TH}_{\lambda=}$ is the syntactic relation $M =_{\beta\eta} N$. For the fixed model K, the proof-relevant layers $\mathrm{Theory}_1(K)$, $\mathrm{Theory}_2(K)$, and $\mathrm{Theory}_3(K)$ enrich $\mathrm{TheoryEq}_K$ by retaining explicit semantic 1-, 2-, and 3-cell witnesses. The global paper-I assertion HoTFT is obtained by quantifying these model-specific layers over all extensional Kan complexes.

Remark 2.5 (Comparing the three theories) The three notions should be kept distinct. The classical theory $\mathrm{TH}_{\lambda=}$ is syntactic and propositional: the assertion $M =_{\beta\eta} N$ records only that some finite conversion chain exists, without remembering any particular chain. For a fixed model K, the relation $\mathrm{TheoryEq}_K$ forgets witnesses in a different way: it asks only that all environments give equal denotations. By contrast, the proof-relevant tower over K retains the *explicit witness*: a pair (M, N) in $\mathrm{Theory}_1(K)$ is accompanied by a specified semantic path, and the higher layers are indexed by parallel lower-dimensional witnesses. The passage from $\mathrm{TH}_{\lambda=}$ to the global HoTFT statement is therefore not merely a set-theoretic strengthening but a genuinely proof-relevant enrichment. This enrichment is not an artefact of the formalism. Theorem 8.7 below demonstrates that the K_∞ model separates β-witnesses from η-witnesses even when they connect the same pair of terms, so the additional structure carried by the proof-relevant tower is genuinely non-degenerate.

2.3. Higher conversions

Definition 2.6 (Explicit higher cells) Given terms M, N, a *2-cell* $\alpha : p \Rightarrow q$ between reduction sequences $p, q : \mathrm{RedSeq}(M, N)$ is a formal witness of homotopy between p and q, built inductively from reflexivity, symmetry, transitivity, congruence under whiskering, horizontal composition, and the structural associator and unitor operations.

A *3-cell* $\Theta : \alpha \Rrightarrow \beta$ between 2-cells is built one dimension higher, with interchange, pentagon, triangle, and the higher whiskering/composition witnesses appearing at this level.

The tower of n-cells is organized into *n-conversions* Σ_n (reduction sequences at dimension 1, homotopies at dimension 2, etc.) and *n-terms* Π_n (explicit witnesses for n-conversions). Here Σ_n denotes the packed nth level of the globular tower; the boundary-indexed fibres $\mathrm{RedSeq}(M, N)$, $\mathrm{Homotopy}_2(p, q)$, and $\mathrm{Homotopy}_3(\alpha, \beta)$ are the corresponding slices over chosen source/target data.

3. Baseline: Soundness and $\mathrm{TH}_{\lambda=} \subseteq \mathrm{HoTFT}$

The results of this section are not new: they were established in Paper I [11] and are restated here in the notation of the present paper so that the subsequent sections can cite them precisely. The reader already familiar with [11] may safely skip to Section 4.

The foundation of the semantic theory is the soundness of β- and η-reduction, and the architecture by which that foundation lifts through the explicit tower is worth stating precisely, both because it fixes notation and because the subsequent sections extend the same inductive strategy to higher-dimensional data.

The logical structure of the lifting is as follows. At the base, one proves that a single β-step (respectively η-step) forces equality of semantic interpretations in any extensional Kan complex; equivalently, it supplies the degenerate semantic 1-cell determined by that equality. This is the content of the two soundness theorems below. A key supporting ingredient is the *substitution lemma*: the semantic interpretation commutes with syntactic substitution,

$$[\![M[N/0]]\!]_\rho \;=\; [\![M]\!]_{\rho[0\mapsto[\![N]\!]_\rho]},$$

which reduces soundness of the β-rule to a computation in the function space. From these individual-step results one passes to composite reduction sequences by induction on sequence length: concatenation of sequences lifts to composition of semantic paths, and formal inversion lifts to symmetry of paths.

At the next level, each inductive constructor of a 2-cell—left whiskering, right whiskering, horizontal composition, associators, and unitors—is shown to lift to a semantic 2-cell, using the composition and whiskering operations of the Kan complex. The 3-cell constructors (interchange, pentagon, triangle, and the higher whiskering/composition witnesses) then lift by the same scheme one dimension higher, using the 3-dimensional horn-filling structure of the Kan complex to produce the required higher homotopies. In each case the inductive step is structural: the semantic image of a composite constructor is the corresponding composite of the semantic images of its constituents.

The three theorems below record the cumulative outcome of this lifting procedure.

Theorem 3.1 (β-soundness) For every extensional Kan complex K, every β-step $M \to_\beta N$ yields equality

$$[\![M]\!]_\rho = [\![N]\!]_\rho$$

for every environment ρ, and hence the corresponding degenerate semantic 1-cell.

Theorem 3.2 (η-soundness) For every extensional Kan complex K, every η-step $M \to_\eta N$ yields equality

$$[\![M]\!]_\rho = [\![N]\!]_\rho$$

for every environment ρ, and hence the corresponding degenerate semantic 1-cell.

By induction on the structure of reduction sequences and their higher-cell constructors, these individual-step soundness results lift to witness-level interpretations of the explicit tower:

Theorem 3.3 ($\mathrm{TH}_{\lambda=} \subseteq \mathrm{Theory}_1(K)$, $\Sigma_2 \subseteq \mathrm{Theory}_2(K)$, $\Sigma_3 \subseteq \mathrm{Theory}_3(K)$) Let K be an extensional Kan complex. Then:

(i) Every reduction sequence $p : \mathrm{RedSeq}(M, N)$ induces a semantic path in K. In particular, whenever $M =_{\beta\eta} N$ in the proposition-level theory $\mathrm{TH}_{\lambda=}$, choosing any explicit reduction-sequence representative yields a proof-relevant semantic lift in $\mathrm{Theory}_1(K)$.

(ii) If $p, q : \mathrm{RedSeq}(M, N)$ and $\alpha : p \Rightarrow q$, then α induces a semantic 2-cell between the interpreted 1-cells of p and q.

(iii) If $p, q : \mathrm{RedSeq}(M, N)$, $\alpha, \beta : p \Rightarrow q$, and $\Theta : \alpha \Rrightarrow \beta$, then Θ induces a semantic 3-cell between the interpreted 2-cells of α and β.

Universal quantification over K recovers the global paper-I theorem $\mathrm{TH}_{\lambda=} \subseteq \mathrm{HoTFT}$ together with its 2- and 3-dimensional companions. These results are exactly those established in [11]. We take them as the starting point for the new theorems presented below.

4. Higher Conversions and the Explicit Tower

4.1. The globular tower of n-conversions

The n-conversion tower is a *globular set*: at dimension 0 one has terms, at dimension 1 reduction sequences between terms, at dimension 2 homotopies between reduction sequences, at dimension 3 higher homotopies between homotopies, and so on. Source and target maps $s, t : \Sigma_{n+1} \to \Sigma_n$ record the boundary of each cell, and the globularity conditions $s \circ s = s \circ t$ and $t \circ s = t \circ t$ hold by construction.

Through dimensions 0–3, the cells are built by explicit inductive types carrying the full combinatorial structure of λ-reduction: whiskering, horizontal composition, associators, pentagon and triangle coherences, interchange, and so on. At dimension 4 and above, the cells are generated by a uniform *recursive completion* principle.

4.2. The low-dimensional coherence core

The low-dimensional structure through dimension 3 forms the core $\mathscr{G}_\lambda$ used later for the recursive completion. Concretely, the following data are available:

- *Composition*: concatenation of reduction sequences.
- *Inverses*: formal reversal of reduction sequences.
- *Whiskering*: left and right whiskering of 2-cells along 1-cells.
- *Horizontal composition*: horizontal composition of 2-cells.
- *Associator*: a 2-cell $\alpha_{p,q,r} : (p \cdot q) \cdot r \Rightarrow p \cdot (q \cdot r)$.
- *Unitors*: 2-cells $\ell_p : \mathrm{id} \cdot p \Rightarrow p$ and $\rho_p : p \cdot \mathrm{id} \Rightarrow p$.
- *Pentagon*: a 3-cell witnessing the standard pentagon identity for four composable 1-cells.
- *Triangle*: a 3-cell witnessing the triangle identity relating the associator and the unitors.
- *Interchange*: 3-cells witnessing both parenthesizations of horizontal composition.

All whiskering operations preserve identity, composition, symmetry, and congruence up to explicit 3-cells.

Definition 4.1 (Low-dimensional λ-coherence core) The core $\mathscr{G}_\lambda$ is the structure $\mathscr{G}_\lambda$ with objects = Term, 1-cells = RedSeq, 2-cells = $\mathrm{Homotopy}_2$, 3-cells = $\mathrm{Homotopy}_3$, and the above operations, satisfying the globularity conditions at all low dimensions.

Definition 4.2 (Reflexive 6-cell extension) For the comparison theorem below we extend the low-dimensional core through dimensions 4, 5, and 6 by adjoining only the formal reflexive–symmetric–transitive layers generated from the 3-cell core:

$$\mathscr{G}^{\leq 6}_{\lambda,4}(\eta,\theta) := \mathrm{HigherDeriv}(\eta,\theta), \qquad \mathscr{G}^{\leq 6}_{\lambda,5}(\omega,\xi) := \mathrm{HigherDeriv}(\omega,\xi),$$

$$\mathscr{G}^{\leq 6}_{\lambda,6}(\mu,\nu) := \mathrm{HigherDeriv}(\mu,\nu),$$

for parallel 3-, 4-, and 5-cells respectively. No new primitive coherence generators are added in these dimensions. We refer to the resulting 0–6 dimensional package as the *reflexive* 6*-cell extension* of $\mathscr{G}_\lambda$.

5. Comparison with the Recursive Completion

5.1. Recursive completion of the tower

This section has one input, one output, and one genuinely non-formal step. The input is the explicit low-dimensional λ-conversion tower together with the reflexive 6-cell extension from Definition 4.2; the output is a realization map from the recursive completion into the explicit all-dimensional tower. The only substantive comparison work occurs in dimensions 4–6, after which the construction is uniform.

Four objects remain distinct throughout what follows: the explicit syntactic tower $\Sigma_\bullet$; the low-dimensional core $\mathscr{G}_\lambda$; its recursive completion $\mathscr{R}_\bullet(\mathscr{G}_\lambda)$; and, later, the semantic towers living in a model K. Theorem 5.6 compares $\mathscr{R}_\bullet(\mathscr{G}_\lambda)$ with $\Sigma_\bullet$; Theorem 6.8 concerns the semantic side.

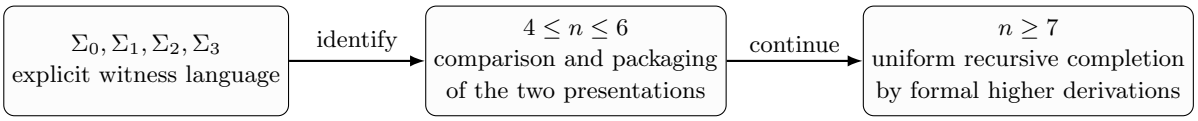
The weak ω-groupoid $\mathscr{G}_\lambda$ constructed from explicit $\beta\eta$-paths already gives concrete data in low dimensions:

$$\Sigma_0 = \mathrm{Term}, \qquad \Sigma_1 = \mathrm{RedSeq}, \qquad \Sigma_2 = \mathrm{Homotopy}_2, \qquad \Sigma_3 = \mathrm{Homotopy}_3.$$

Three layers are in play throughout this section: the primitive syntactic tower Σ_0–Σ_3; the reflexive 6-cell extension used by the generic coherence API; and the recursive completion that continues uniformly above dimension 6. The point of the argument is to identify these layers where they overlap and then pass to the common recursive continuation.

Moreover, the low-dimensional core also carries the next layers of reflexive higher coherence needed to iterate the construction. The point of this section is therefore structural rather than model-specific: once this finite core is fixed, no new primitive coherence operations are needed in higher dimension, and one can continue the tower uniformly by freely adding formal reflexivity, symmetry, and transitivity witnesses. The non-formal part of the argument is to show that this equality-generated continuation agrees with the explicit low-dimensional presentation and preserves strict globular boundaries. Figure 1 summarizes the three phases of that comparison.

Remark 5.1 (Which layers are explicit) Two different presentations are in play here. The syntactic tower $\Sigma_\bullet$ is explicit through dimension 3, and for $n \geq 4$ its cells are already given recursively as higher-derivation data between parallel $(n-1)$-cells. Separately, the shared simplicial core used by the generic coherence packaging is a reflexive 6-cell core: its 4-, 5-, and 6-cells are packaged reflexivity layers attached to the low-dimensional λ-coherence interface. The role of dimensions 4–6 in Theorem 5.6 is precisely to identify these two presentations before the uniform recursive continuation begins above dimension 6.



$\mathscr{R}_\bullet(\mathscr{G}_\lambda)$ admits a strict boundary-preserving realization
inside the explicit tower $\Sigma_\bullet$.

Figure 1 Architecture of the proof of Theorem 5.6. The explicit low-dimensional tower is first matched with the equality-generated continuation in the bridge dimensions, after which the recursive completion proceeds uniformly.

Definition 5.2 (Higher derivation) Let X be a type. The type $\mathrm{HigherDeriv}_X(x, y)$ of *higher derivations* between $x, y \in X$ is generated inductively by

$$\mathsf{refl}_x : \mathrm{HigherDeriv}_X(x, x), \qquad \mathsf{symm} : \mathrm{HigherDeriv}_X(x, y) \to \mathrm{HigherDeriv}_X(y, x),$$

and

$$\mathsf{trans} : \mathrm{HigherDeriv}_X(x, y) \to \mathrm{HigherDeriv}_X(y, z) \to \mathrm{HigherDeriv}_X(x, z).$$

Thus $\mathrm{HigherDeriv}_X$ is the reflexive–symmetric–transitive closure of the equality data already present on X.

The type $\mathrm{HigherDeriv}_X(x, y)$ does not introduce new primitive generators between unrelated objects. In particular, any element of $\mathrm{HigherDeriv}_X(x, y)$ determines that the endpoints agree. Accordingly, above the explicit core the recursive tower is *equality-generated*: higher cells are added by iterating formal reflexivity, symmetry, and transitivity on previously constructed parallel cells. Its central structural property is still *functoriality*: every map on the underlying type transports higher derivations covariantly.

Proposition 5.3 (Functoriality of HigherDeriv) Let $f : X \to Y$ be any map. There is a canonical map

$$\mathrm{HigherDeriv}(f) : \mathrm{HigherDeriv}_X(x, y) \longrightarrow \mathrm{HigherDeriv}_Y\big(f(x),\, f(y)\big),$$

defined by recursion on the constructors of HigherDeriv:

$$\begin{aligned}
\mathrm{HigherDeriv}(f)(\mathsf{refl}_x) &= \mathsf{refl}_{f(x)}, \\
\mathrm{HigherDeriv}(f)(\mathsf{symm}(h)) &= \mathsf{symm}\big(\mathrm{HigherDeriv}(f)(h)\big), \\
\mathrm{HigherDeriv}(f)(\mathsf{trans}(h_1, h_2)) &= \mathsf{trans}\big(\mathrm{HigherDeriv}(f)(h_1),\, \mathrm{HigherDeriv}(f)(h_2)\big).
\end{aligned}$$

Moreover, the assignment $f \mapsto \mathrm{HigherDeriv}(f)$ respects identity and composition:

$$\mathrm{HigherDeriv}(\mathrm{id}_X) = \mathrm{id}, \qquad \mathrm{HigherDeriv}(g \circ f) = \mathrm{HigherDeriv}(g) \circ \mathrm{HigherDeriv}(f).$$

Proof The recursive clauses define $\mathrm{HigherDeriv}(f)$ by structural induction on derivation terms. The identity law is immediate: $\mathrm{HigherDeriv}(\mathrm{id})$ fixes every constructor pointwise, since $\mathsf{refl}_x \mapsto \mathsf{refl}_{\mathrm{id}(x)} = \mathsf{refl}_x$, $\mathsf{symm}(h) \mapsto \mathsf{symm}(\mathrm{HigherDeriv}(\mathrm{id})(h))$, and the induction hypothesis closes the recursion.

For the composition law, the base case is:

$$\begin{aligned}
\mathrm{HigherDeriv}(g)(\mathrm{HigherDeriv}(f)(\mathsf{refl}_x)) &= \mathrm{HigherDeriv}(g)(\mathsf{refl}_{f(x)}) \\
&= \mathsf{refl}_{g(f(x))} \\
&= \mathrm{HigherDeriv}(g \circ f)(\mathsf{refl}_x).
\end{aligned}$$

The inductive cases for symm and trans follow by the same structural unwinding: the outer map $\mathrm{HigherDeriv}(g)$ distributes across the constructors produced by $\mathrm{HigherDeriv}(f)$, and the induction hypothesis matches each sub-derivation. □

Proposition 5.3 is the engine that drives the realization theorem below: at each dimension, the realization of a higher derivation is obtained by applying HigherDeriv to the endpoint realization map. The identity and composition laws ensure that this process is compatible with the globular boundary structure and with the recursive definition of source and target.

Starting from the reflexive 6-cell extension of $\mathscr{G}_\lambda$, we define a recursive globular tower $\mathscr{R}_\bullet(\mathscr{G}_\lambda)$ as follows. For dimensions $0, \ldots, 6$ we keep the given low-dimensional cells of the core explicitly. For every $n \geq 6$, an $(n+1)$-cell is a triple

$$(x, y, h), \qquad x, y \in \mathscr{R}_n(\mathscr{G}_\lambda), \quad h \in \mathrm{HigherDeriv}(x, y).$$

Its source is x and its target is y. The globularity equalities are then automatic: if h is a higher derivation from x to y, the two iterated sources and the two iterated targets agree because every constructor of HigherDeriv preserves equality of endpoints.

This recursive completion should be compared with the explicit tower $\Sigma_\bullet$ of higher λ-conversions. The latter is already constructive in all dimensions, but its presentation is not literally the same as the generic recursive completion in dimensions $4, 5, 6$: there one must package the previously constructed 3-cells, 4-cells, and 5-cells as explicit cells of the target tower. Once these dimensions are handled, however, the higher stages on both sides are governed by the same rule: "a higher cell is a higher derivation between parallel cells of the preceding dimension."

The interface between the explicit low-dimensional tower and the recursive completion in the intermediate dimensions is captured by the following observation, which we state as a separate lemma because it isolates the only non-trivial step in passing from the low-dimensional core to the recursive continuation.

Lemma 5.4 (Low-dimensional packaging) For $d \in \{3, 4, 5\}$, let Σ_d denote the explicit d-cells of the λ-ω-groupoid, and let $\mathscr{R}_{d+1}(\mathscr{G}_\lambda)$ denote the $(d+1)$-cells of the recursive completion. There is a canonical *packaging map*

$$\mathsf{pack}_{d+1} : \mathscr{R}_{d+1}(\mathscr{G}_\lambda) \longrightarrow \Sigma_{d+1}$$

defined by:

(a) embedding the two endpoint d-cells into Σ_d via the identity (for $d \leq 3$) or via the inductively defined packaging map pack_d (for $d \in \{4, 5\}$), and then
(b) transporting the higher derivation datum along the resulting endpoint map, using the functoriality of HigherDeriv (Proposition 5.3).

No new primitive coherence operations beyond those already present in the 3-cell core enter the construction.

Proof At dimension 4, the endpoints are 3-cells, which belong to the explicit core and therefore need no packaging: pack_4 applies HigherDeriv(id) to the derivation datum, which by the identity law of Proposition 5.3 is the identity. Concretely, a 4-cell of the recursive completion consists of two parallel 3-cells η, θ together with a higher derivation $\omega : \mathrm{HigherDeriv}(\eta, \theta)$, and pack_4 sends this to the explicit 4-cell (η, θ, ω) verbatim.

At dimension 5, the endpoints are 4-cells. Each such 4-cell was itself produced by pack_4, so pack_5 applies $\mathrm{HigherDeriv}(\mathsf{pack}_4)$ to the derivation data. The composition law of Proposition 5.3 guarantees that iterated packaging respects the boundary maps. The argument for dimension 6 is identical, with pack_5 replacing pack_4.

The key observation is that every constructor invoked—refl, symm, trans, and the endpoint maps—already exists in the 3-cell core. The packaging maps merely *compose* these existing operations through the functoriality of HigherDeriv; they do not introduce any new generating data. □

5.2. The realization theorem

Definition 5.5 (Boundary-preserving realization package) Let $\mathscr{C}$ be a reflexive low-dimensional globular core and let $\mathscr{T}$ be a globular tower. A *boundary-preserving realization package* from $\mathscr{C}$ to $\mathscr{T}$ consists of maps

$$\mathrm{realize}_n : \mathscr{R}_n(\mathscr{C}) \longrightarrow \mathscr{T}_n \qquad (n \in \mathbb{N}),$$

where $\mathscr{R}_\bullet(\mathscr{C})$ is the recursive completion of $\mathscr{C}$, such that for every $(n+1)$-cell x,

$$s(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(sx), \qquad t(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(tx).$$

Theorem 5.6 (Comparison with the recursive completion) The explicit higher λ-conversion tower $\Sigma_\bullet$ admits an boundary-preserving realization package from the recursive completion of the low-dimensional core $\mathscr{G}_\lambda$. In particular:

(i) For every $n \in \mathbb{N}$, there is a realization map
$$\mathrm{realize}_n : \mathscr{R}_n(\mathscr{G}_\lambda) \to \Sigma_n.$$
(ii) These realization maps commute strictly with source and target maps.
(iii) Hence $\Sigma_\bullet$ carries a strict boundary-preserving realization of $\mathscr{R}_\bullet(\mathscr{G}_\lambda)$ extending the low-dimensional core.

Proof We divide the construction into the explicit core, the finite packaging phase, and the uniform recursive continuation.

Step 1: dimensions 0–3. At the first four levels there is nothing to prove beyond unpacking the definitions. The objects of $\mathscr{G}_\lambda$ are λ-terms, its 1-cells are explicit reduction zigzags, its 2-cells are explicit homotopies between parallel zigzags, and its 3-cells are explicit homotopies between parallel 2-cells. Thus

$$\mathrm{realize}_0, \mathrm{realize}_1, \mathrm{realize}_2, \mathrm{realize}_3$$

are simply the identity identifications between the recursively completed tower and the explicit tower.

Step 2: dimensions 4–6. By Definition 4.2, the intermediate dimensions are already the formal reflexive layers generated from the 3-cell core. Lemma 5.4 therefore provides canonical packaging maps

$$\mathsf{pack}_4, \qquad \mathsf{pack}_5, \qquad \mathsf{pack}_6$$

from the recursive side to the corresponding explicit cells of $\Sigma_4, \Sigma_5, \Sigma_6$. We define

$$\mathrm{realize}_4 := \mathsf{pack}_4, \qquad \mathrm{realize}_5 := \mathsf{pack}_5, \qquad \mathrm{realize}_6 := \mathsf{pack}_6.$$

The source/target commutation in these dimensions is exactly the content of Lemma 5.4 together with the functoriality of HigherDeriv from Proposition 5.3: packaging an endpoint first and then taking its boundary agrees with taking the boundary in the reflexive 6-cell extension first and then packaging the result.

Step 3: recursive continuation above dimension 6. Suppose now that $\mathrm{realize}_n$ has already been defined for some $n \geq 6$. An $(n+1)$-cell of the recursive completion is, by definition, a triple (x, y, h) with $x, y \in \mathscr{R}_n(\mathscr{G}_\lambda)$ and $h \in \mathrm{HigherDeriv}(x, y)$. We define

$$\mathrm{realize}_{n+1}(x, y, h) := \big(\mathrm{realize}_n(x),\ \mathrm{realize}_n(y),\ \mathrm{HigherDeriv}(\mathrm{realize}_n)(h)\big).$$

Here $\mathrm{HigherDeriv}(\mathrm{realize}_n)(h)$ denotes the image of h under the map obtained by recursion on the constructors of HigherDeriv:

$$\mathsf{refl}(x) \mapsto \mathsf{refl}(\mathrm{realize}_n(x)), \qquad \mathsf{symm}(h) \mapsto \mathsf{symm}(\mathrm{HigherDeriv}(\mathrm{realize}_n)(h)),$$

$$\mathsf{trans}(h_1, h_2) \mapsto \mathsf{trans}(\mathrm{HigherDeriv}(\mathrm{realize}_n)(h_1), \mathrm{HigherDeriv}(\mathrm{realize}_n)(h_2)).$$

Thus the realization in dimension $n+1$ is forced by the realization in dimension n.

The conceptual point is worth highlighting: no new coherence operations, packaging maps, or case analyses enter the construction above dimension 6. The reason is structural. Below dimension 4, the explicit tower is defined by a bespoke inductive type carrying whiskering, interchange, pentagon, and triangle constructors. In dimensions 4–6, one must translate between that bespoke presentation and the generic recursive completion, which requires the packaging maps of Lemma 5.4. But from dimension 7 onward, *both* towers are defined by the same rule: a cell is a higher derivation between parallel cells of the preceding dimension. Above that range, the realization map becomes uniform: it is given by the same higher-derivation recursion on both sides, so no new comparison data are needed beyond the inductive hypothesis. In particular, no finite list of "exceptional dimensions" grows as one climbs the tower: the base and packaging phases are completed once and for all in dimensions 0–6, and the rest is pure induction.

Step 4: compatibility with source and target. For dimensions 0–6, compatibility is checked directly from the formulas: both the recursive tower and the explicit λ-tower compute source by forgetting the last boundary component, and compute target by forgetting the corresponding source component.

For dimensions above 6, let $x = (u, v, h)$ be an $(n+1)$-cell. Then, by definition of the recursive source and target maps,

$$s(x) = u, \qquad t(x) = v.$$

Applying the recursive formula for realization gives

$$\mathrm{realize}_{n+1}(x) = \big(\mathrm{realize}_n(u), \mathrm{realize}_n(v), \mathrm{HigherDeriv}(\mathrm{realize}_n)(h)\big).$$

Hence its source in the explicit tower is exactly $\mathrm{realize}_n(u)$ and its target is exactly $\mathrm{realize}_n(v)$, so

$$s(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(sx), \qquad t(\mathrm{realize}_{n+1}(x)) = \mathrm{realize}_n(tx).$$

This proves strict source and target compatibility in all dimensions.

Step 5: conclusion. The realization maps realize the recursively completed tower of the λ-coherence core inside the explicit higher λ-conversion tower at every dimension. Since the recursive completion is globular by construction and the realization preserves boundaries strictly, the explicit tower inherits the boundary-preserving realization package of the theorem statement. □

5.3. Truncation and the classical theory

An immediate corollary of Theorem 5.6 is that the recursive completion extends, rather than replaces, the ordinary theory of $\beta\eta$-equality.

Corollary 5.7 (0-truncation) The 0-truncation of the low-dimensional core—and therefore also of its recursive completion—recovers the classical $\beta\eta$-equality:

$$\pi_0(\mathscr{G}_\lambda)(M, N) \iff M =_{\beta\eta} N.$$

Proof The proposition $\pi_0(\mathscr{G}_\lambda)(M, N)$ asserts only that there exists a 1-cell from M to N, that is, an explicit reduction sequence $p : \mathrm{RedSeq}(M, N)$. Forgetting the witness p yields an ordinary $\beta\eta$-conversion from M to N. Conversely, every ordinary $\beta\eta$-conversion admits an explicit reduction-sequence representative. Thus existence of a 1-cell in the recursive completion—and hence in its realization inside the explicit tower—is equivalent to the classical proposition-level equality relation. □

6. Sufficient Semantic Coherence Data

This section asks a narrow semantic question: how much chosen coherence data are actually needed for the later associator and pentagon arguments? The input is an extensional Kan complex equipped with a specific small front-facing package of 3-cells; the output is the recursive associator comparison, the semantic pentagon comparison on interpreted reduction sequences, and the bridge theorems used later.

6.1. Three semantic regimes

The semantic development separates three logically distinct layers.

(1) **Bare semantics.** An extensional Kan complex suffices for the interpretation of terms, β- and η-soundness, the interpretation of explicit reduction sequences, and the semantic images of the explicit 2- and 3-cells already present in the syntax.

(2) **Coherent semantics.** A coherent extensional Kan complex adds explicit 3-dimensional comparison data for associativity and whiskering. This makes it possible to compare different semantic parenthesizations of composite interpreted paths.

(3) **Strict semantics.** In a strict extensional Kan complex, the required coherent comparison data are derived axiom-freely from uniqueness of horn fillers, so no additional chosen coherence fields are needed beyond the extensional λ-model structure.

We now show that a smaller front-facing seed already generates the full associator and pentagon package needed later.

6.2. The front-seed boundary data

Definition 6.1 (Front-seed coherent extensional Kan complex) A *front-seed coherent extensional Kan complex* is an extensional Kan complex K equipped with the following additional 3-cell data.

(FS1) **WLWR comparison.** For every composable 1-cells

$$a \xrightarrow{\alpha} b \xrightarrow{\beta,\gamma} c \xrightarrow{\delta} d$$

and every 2-cell $\eta : \beta \Rightarrow \gamma$, a chosen 3-cell

$$\mathrm{wr}(\mathrm{wl}(\alpha, \eta), \delta) \implies \mathrm{ass}(\alpha, \beta, \delta) \cdot \mathrm{wl}(\alpha, \mathrm{wr}(\eta, \delta)) \cdot \mathrm{ass}(\alpha, \gamma, \delta)^{-1}.$$

In words: right-whiskering a left-whiskered 2-cell agrees, up to the appropriate associator corrections, with first right-whiskering and then left-whiskering.

(FS2) **Inner-right-front pentagon contraction.** For every composable quadruple p, q, r, s, a chosen 3-cell contracting the inner-right-front face of the semantic pentagon to the reflexive 2-cell on the edge labeled by $q \cdot (r \cdot s)$ before left-whiskering by p.

The second datum is deliberately asymmetric: it does not assert the entire pentagon at once, but only contracts one distinguished front face. The theorem below says that, together with WLWR, this already forces the higher coherence needed for associator comparison and for the full pentagon at interpreted reduction sequences.

It is useful to describe the geometric content of the two seeds more precisely.

FS1 (WLWR comparison): front-face horn filling for mixed whiskering. Given a 2-cell $\eta : \beta \Rightarrow \gamma$ flanked by 1-cells α on the left and δ on the right, there are two natural ways to extend η to the full composite boundary: first left-whisker by α and then right-whisker by δ, or first right-whisker by δ and then left-whisker by α. These two expressions land at the same pair of endpoints $\alpha \cdot \beta \cdot \delta$ and $\alpha \cdot \gamma \cdot \delta$, but they are not *a priori* the same 2-cell because the whiskering operations interact

with the associativity of path composition. Seed (FS1) asserts that the discrepancy is controlled by the associator: the two mixed whiskerings differ by exactly the pair of associator 2-cells that rebrackets the composite. In simplicial terms, FS1 fills a specific 3-horn whose two front faces are the two mixed-whiskering composites and whose remaining faces are associator instances.

FS2 (inner-right-front pentagon contraction): partial boundary seeding. The pentagon coherence for four composable 1-cells p, q, r, s is a 3-cell whose boundary consists of five associator 2-cells arranged around the boundary of a 4-simplex. Rather than positing the full pentagon 3-cell as primitive, FS2 provides only the contraction of the *inner-right-front* face—the face built from the associator $\mathrm{ass}(q, r, s)$ whiskered by p on the left. This contracts a particular sub-composite of the pentagon boundary to the identity 2-cell on the edge labeled by $q \cdot (r \cdot s)$ before left-whiskering by p. Geometrically, FS2 is a partial horn-filling datum: it specifies one non-degenerate face of the pentagon 4-horn, from which the remaining faces can be assembled using the WLWR comparison of FS1 together with the standard Kan-complex horn-filling machinery.

Definition 6.2 (Structural pentagon data) For composable reduction sequences p, q, r, s, write

$$P_0 := (((p \cdot q) \cdot r) \cdot s), \qquad P_1 := ((p \cdot q) \cdot (r \cdot s)), \qquad P_2 := ((p \cdot (q \cdot r)) \cdot s),$$

$$P_3 := p \cdot ((q \cdot r) \cdot s), \qquad P_4 := p \cdot (q \cdot (r \cdot s)).$$

The two 2-dimensional boundary composites of the structural pentagon are

$$L_{p,q,r,s} := \mathrm{ass}(p \cdot q, r, s) \ \cdot \ \mathrm{ass}(p, q, r \cdot s) \quad \text{and} \quad R_{p,q,r,s} := (\mathrm{ass}(p, q, r) * s) \ \cdot \ \mathrm{ass}(p, q \cdot r, s) \ \cdot \ (p * \mathrm{ass}(q, r, s)),$$

both regarded as 2-cells from P_0 to P_4. We call the partial 4-dimensional boundary built from the FS2 inner-right-front contraction, the two WLWR comparison faces, and the remaining back associator face the *structural pentagon horn* on p, q, r, s; its missing face is the desired 3-cell $L_{p,q,r,s} \Rightarrow R_{p,q,r,s}$.

The significance of Theorem 6.8 is a precise reduction of the semantic coherence package actually used later. In the formal development, the larger coherent interface with a full pentagon witness and the smaller front-seed interface are both kept explicitly. The front-seed theorem shows that the smaller chosen-data interface already suffices: the full pentagon field may be replaced by the inner-right-front contraction, from which one derives the recursive associator comparison, the structural semantic pentagon comparison on interpreted reduction sequences, and the source/target/shell bridge theorems for the explicit syntactic pentagon, with the target and shell bridges landing in the mixed target shell determined by the front-seed associator comparison. Thus the gain is not an appeal to an unspecified larger coherence package, but a concrete reduction from a full pentagon datum to a smaller front-facing seed, while keeping WLWR explicit.

Lemma 6.3 (WLWR head-step normalization) Let K be a front-seed coherent extensional Kan complex, let α be the semantic image of a generating head step, let $\eta : \beta \Rightarrow \gamma$ be a semantic 2-cell, and let δ be a further semantic edge. Then the mixed whiskering composite

$$\mathrm{wr}(\mathrm{wl}(\alpha, \eta), \delta)$$

is connected by a chosen semantic 3-cell to

$$\mathrm{ass}(\alpha, \beta, \delta) \ \cdot \ \mathrm{wl}(\alpha, \mathrm{wr}(\eta, \delta)) \ \cdot \ \mathrm{ass}(\alpha, \gamma, \delta)^{-1}.$$

Proof This is exactly the WLWR comparison supplied by Seed (FS1), recorded here as a standalone normalization lemma because it is the only genuinely non-formal head-step input used in the associator and pentagon arguments below. □

Lemma 6.4 (Triple associator comparison) Let K be a front-seed coherent extensional Kan complex and let p, q, r be reduction sequences. Then the structural associator shell of the interpreted composites $((p \cdot q) \cdot r)$ and $p \cdot (q \cdot r)$ is connected by a semantic 3-cell to the equality-generated associator comparison coming from literal concatenation.

Proof Proceed by induction on the first reduction sequence p.

Base case. If $p = \mathsf{refl}$, then the structural and equality-generated associator comparisons coincide, so the required 3-cell is reflexive.

Head-step case. Suppose p begins with a generating step whose semantic image is α, and write the remaining tail comparison as $\eta : \beta \Rightarrow \gamma$. The only non-formal obstruction is the nested whiskering term

$$\mathrm{wr}(\mathrm{wl}(\alpha, \eta), \delta),$$

where δ is the semantic image of r. By Lemma 6.3, this term is replaced by the normalized composite

$$\mathrm{ass}(\alpha, \beta, \delta) \ \cdot \ \mathrm{wl}(\alpha, \mathrm{wr}(\eta, \delta)) \ \cdot \ \mathrm{ass}(\alpha, \gamma, \delta)^{-1}.$$

After this explicit head-step normalization, the boundary matches the structural associator shell.

Inductive step. Write p as its head step followed by a shorter tail. Apply the WLWR normalization above to the head step and the induction hypothesis to the tail. Vertical composition of the resulting 3-cells produces a comparison from the structural associator shell of (p, q, r) to the equality-generated associator. □

Lemma 6.5 (Front-seed pentagon horn completion) Let K be a front-seed coherent extensional Kan complex and let p, q, r, s be reduction sequences. Then the structural pentagon horn of Definition 6.2 is a well-defined 4-dimensional horn in K.

Proof The four prescribed faces are:

(a) the inner-right-front face supplied by Seed (FS2),
(b) the two front mixed-whiskering faces normalized by Seed (FS1),
(c) the remaining back face built from the structural associator shell.

The inner-right-front face has target the reflexive 2-cell on P_4. Each front WLWR face has as one boundary edge the same mixed-whiskering composite as the adjacent FS2 or associator face, and Lemma 6.4 identifies the remaining boundary edges of those WLWR faces with the corresponding associator composites built from $L_{p,q,r,s}$ and $R_{p,q,r,s}$. The back face is already the structural associator shell on the same parenthesization data. Consequently every common 2-dimensional overlap is described by the same composite of associators and whiskerings when read from either adjacent 3-face. Hence the four prescribed faces glue to a well-defined 4-horn in K. □

Lemma 6.6 (Source-side associator bridge) Let K be a front-seed coherent extensional Kan complex and let p, q, r, s be reduction sequences. The interpreted source boundary of the explicit syntactic pentagon is connected by a semantic 3-cell to the structural semantic source shell determined by $L_{p,q,r,s}$.

Proof Apply Lemma 6.4 to the two associator pieces on the source side of the explicit pentagon. Their composite identifies the interpreted syntactic source boundary with the structural source shell. □

Lemma 6.7 (Target-side associator bridge) Let K be a front-seed coherent extensional Kan complex and let p, q, r, s be reduction sequences. The interpreted target boundary of the explicit syntactic pentagon is connected by a semantic 3-cell to the mixed target shell determined by $R_{p,q,r,s}$.

Proof Apply Lemma 6.4 to the three associator pieces on the target side of the explicit pentagon. Their composite identifies the interpreted syntactic target boundary with the mixed target shell. □

Theorem 6.8 (Front-seed sufficiency) Let K be a front-seed coherent extensional Kan complex. Then:

(i) For every triple of reduction sequences p, q, r, there is a semantic 3-cell from the structural associator shell of the interpreted composites $((p \cdot q) \cdot r)$ and $p \cdot (q \cdot r)$ to the equality-generated associator comparison coming from literal concatenation. Consequently any two parenthesizations of a finite composite are connected by a chain of such comparison 3-cells.
(ii) For every quadruple of reduction sequences p, q, r, s, there is a semantic 3-cell

$$\mathrm{ass}(p \cdot q, r, s) \ \cdot \ \mathrm{ass}(p, q, r \cdot s) \implies (\mathrm{ass}(p, q, r) * s) \ \cdot \ \mathrm{ass}(p, q \cdot r, s) \ \cdot \ (p * \mathrm{ass}(q, r, s)),$$

where $*$ denotes whiskering.
(iii) For every quadruple p, q, r, s, the interpreted explicit syntactic pentagon admits a source bridge to the structural semantic source shell, a target bridge to the mixed target shell determined by (i), and the resulting shell bridge between those two shells.

Clause (ii) gives the fully structural semantic pentagon comparison, whereas clause (iii) bridges the explicit syntactic pentagon only to the mixed target shell arising from the front-seed associator comparison.

Proof We prove the three claims in order.

Step 1: from WLWR to the recursive associator comparison. Lemma 6.4 gives the comparison for every triple p, q, r. Iterating that comparison gives the statement for arbitrary parenthesized composites of any finite list $p_1, \ldots, p_k$: one changes parenthesization one associator at a time, and each elementary move is controlled by the triple comparison lemma.

Step 2: from the front pentagon seed to the full semantic pentagon. By Lemma 6.5, the four known faces assemble to the structural pentagon horn of Definition 6.2. In the underlying simplicial set of K, this boundary is represented by the specific $(2, 1)$-horn used in the formal proof: face 0 is the FS2 inner-right-front contraction, face 2 is the composition triangle for p against the already parenthesized tail $q \cdot (r \cdot s)$, and face 3 is the composite of the two FS1-normalized mixed-whiskering faces with the back associator shell. The missing face 1 is therefore exactly the desired pentagon comparison $L_{p,q,r,s} \Rightarrow R_{p,q,r,s}$, and the Kan filling property supplies it.

Step 3: bridge theorems from syntax to semantic shells. The explicit syntactic higher calculus already contains an associator 2-cell and a pentagon 3-cell between explicit reduction sequences. These objects can be interpreted in any bare extensional Kan complex, so in particular in our front-seed model K. However, their semantic images land a priori in the "syntactic" presentation of the boundary, while Step 1 and Step 2 produce the structural source shell and the mixed target shell built from the Kan-complex associator and whiskering operations.

Lemma 6.6 supplies the source-side bridge and Lemma 6.7 supplies the target-side bridge. Finally, compose

(a) the source-side bridge,
(b) the interpreted syntactic pentagon 3-cell, and
(c) the inverse of the target-side bridge.

The result is a 3-cell from the structural source shell to the mixed target shell. This is exactly the front-seed shell bridge theorem. Thus the explicit syntactic pentagon and the front-seed semantic pentagon boundary are not merely parallel facts: they are connected by explicit comparison 3-cells, while Step 2 separately provides the full structural semantic pentagon comparison on interpreted reduction sequences.

This proves (i), (ii), and (iii). □

Remark 6.9 (Open strengthening target) The remaining conceptual question is whether the two front-seed data can be derived from the bare extensional Kan interface alone. A positive answer would show that no additional semantic coherence fields are needed even for the associator and pentagon package. At present, what is proved is sharper but more modest: the full coherent interface is unnecessary, and one specific smaller boundary package is already sufficient, namely the WLWR comparison together with the inner-right-front pentagon contraction.

7. The K_∞ Model

The K_∞ model was introduced in Paper II [10], which defined the inverse-limit construction, proved algebraicity and bounded completeness (Proposition 4.1), and exhibited the separation phenomenon of Example 8.1. The aspects of that baseline used here are restated in this section and in Section 8. The main new point of the present section is not the abstract fact that a projection-pair inverse limit can be reflexive—a classical theme already present in Scott-style semantics and later domain-theoretic accounts [13, 1]—but the *exact reflexive packaging*: explicit, globally continuous reify and reflect maps together with a stagewise application formula that pins down the relationship between the finite stages and the limit.

7.1. Construction

The object K_∞ is obtained as the inverse limit of the reflexive tower

$$K_0,\ K_1,\ K_2,\ \ldots, \qquad K_0 = N^+, \qquad K_{n+1} = [K_n \to K_n].$$

At each stage there is a projection pair

$$f_n^+ : K_n \longrightarrow K_{n+1}, \qquad f_n^- : K_{n+1} \longrightarrow K_n$$

with

$$f_n^- \circ f_n^+ = \mathrm{id}_{K_n}, \qquad f_n^+\big(f_n^-(u)\big) \le u \quad (u \in K_{n+1}).$$

Thus every successor stage is a reflexive enlargement of the previous one.

Here and below, $u \simeq v$ means mutual approximation $u \le v$ and $v \le u$ in the underlying homotopy partial order. We use $\simeq$ only for inverse-limit coherence and density statements; the exact formulas in Theorem 7.15 are literal equalities of continuous maps and coordinates.

Definition 7.1 (The tower $K_0, K_1, \ldots$) Set $K_0 = N^+$, the flat domain on the non-trivial homotopy-group carrier $N = \bigsqcup_{n\geq 0} S^n$ together with bottom, and define inductively

$$K_{n+1} = [K_n \to K_n].$$

The transition maps are the projection pairs (f_n^+, f_n^-) above.

Definition 7.2 (K_∞) The inverse limit K_∞ is the set of coherent threads

$$K_\infty = \varprojlim_n K_n = \Big\{(x_n)_{n\geq 0} \in \prod_{n\geq 0} K_n : f_n^-(x_{n+1}) \simeq x_n \text{ for all } n\Big\},$$

ordered coordinatewise. We write

$$\pi_n : K_\infty \to K_n$$

for the nth projection.

Definition 7.3 (Canonical stage embedding) For $u \in K_n$, define the thread $f_{n,\infty}(u) \in K_\infty$ by its coordinates

$$\pi_m(f_{n,\infty}(u)) = \begin{cases} f_m^- \circ \cdots \circ f_{n-1}^-(u), & m < n, \\ u, & m = n, \\ f_{m-1}^+ \circ \cdots \circ f_n^+(u), & m > n. \end{cases}$$

Lemma 7.4 (Coherence of the canonical stage embedding) The thread of Definition 7.3 is coherent. Moreover

$$\pi_n \circ f_{n,\infty} = \mathrm{id}_{K_n}, \qquad f_{n+1,\infty} \circ f_n^+ = f_{n,\infty}.$$

Proof For $m < n$, the defining coordinate formulas satisfy

$$f_m^-\big(\pi_{m+1}(f_{n,\infty}(u))\big) = f_m^-\big(f_{m+1}^- \circ \cdots \circ f_{n-1}^-(u)\big) = \pi_m(f_{n,\infty}(u)),$$

while for $m \geq n$ the coherence is exactly the projection-pair equation $f_m^- f_m^+ = \mathrm{id}$. Thus the coordinates form a coherent thread. The identity $\pi_n \circ f_{n,\infty} = \mathrm{id}_{K_n}$ is immediate from the middle clause of the definition, and the compatibility $f_{n+1,\infty} \circ f_n^+ = f_{n,\infty}$ follows by comparing the two sides coordinatewise. □

Remark 7.5 (Thread extensionality) Thread equality in K_∞ is coordinatewise: two elements of K_∞ are equal iff all of their projections π_n agree. We use this extensionality repeatedly below. By contrast, the relation $\simeq$ records only mutual approximation in the homotopy order and is used to state coherence and density before exact equalities are extracted.

Remark 7.6 (The canonical higher tower on K_∞) The higher-cell tower used for K_∞ in Section 8 is the canonical identity-type ω-groupoid on the carrier of K_∞: a 1-cell from x to y is an equality $x = y$, a 2-cell is an equality between such equalities, and higher cells are iterated equalities. This choice adds no extra K_∞-specific higher-conversion axioms and is the specific tower studied in the present paper. Consequently, once two points of K_∞ are distinct, they admit no 1-cell and hence no higher cell in this particular tower. Theorem 8.7 should therefore be read as a fixed-span persistence statement for the canonical tower, not as a classification of all possible higher structures that one might place on K_∞.

The domain-theoretic content of the construction is the following.

Lemma 7.7 (Finite-stage compact factorization) Let $x \in K_\infty$ and let $c \ll x$ be compact-below. Then there exist $n \geq 0$ and $b \ll \pi_n(x)$ in K_n such that

$$c \leq f_{n,\infty}(b) \leq f_{n,\infty}(\pi_n(x)).$$

Proof Compact-below information in the inverse limit is detected at a finite stage: because c is compact, only finitely many coordinates are needed to witness the inequality $c \leq x$. Choose n larger than that finite support and let $b := \pi_n(c)$. Then $b \ll \pi_n(x)$ in the algebraic stage K_n, and the coordinate formula of Definition 7.3 shows that the embedded stage-n thread $f_{n,\infty}(b)$ dominates c on every coordinate while still lying below the stage-n approximant $f_{n,\infty}(\pi_n(x))$. □

Lemma 7.8 (Coordinatewise least upper bounds are coherent) Let $X \subseteq K_\infty$ be upper-bounded and set

$$u_n := \sup \pi_n(X) \in K_n.$$

Then

$$f_n^-(u_{n+1}) = u_n \qquad \text{for all } n \geq 0.$$

Consequently $(u_n)_{n\geq 0}$ is a coherent thread in K_∞ and is the least upper bound of X.

Proof For every $x \in X$, thread coherence gives

$$f_n^-(\pi_{n+1}(x)) = \pi_n(x).$$

Since f_n^- is monotone and preserves least upper bounds of upper-bounded families, applying it to $u_{n+1} = \sup \pi_{n+1}(X)$ yields an upper bound of $\pi_n(X)$. Conversely, every upper bound of $\pi_n(X)$ pulls back along f_n^- to an upper bound of $\pi_{n+1}(X)$ by the projection-pair equations, so leastness forces $f_n^-(u_{n+1}) = u_n$. The thread condition follows, and coordinatewise leastness then shows that $(u_n)_n$ is $\sup X$ in K_∞. □

Proposition 7.9 (Proposition 4.1) K_∞ is a homotopy Scott domain: it is algebraic and bounded complete.

Proof We first analyze the finite stages.

Bounded completeness of the stages. The base $K_0 = N^+$ is a flat domain, hence bounded complete. If K_n is bounded complete, then the function space $K_{n+1} = [K_n \to K_n]$ is bounded complete under the pointwise order, since least upper bounds of upper-bounded families are computed pointwise. Therefore every K_n is bounded complete.

Algebraicity of the stages. The base stage K_0 is algebraic: every non-bottom element is compact, and every element is the least upper bound of the compact elements below it. Assume now that K_n is algebraic. We show that $K_{n+1} = [K_n \to K_n]$ is algebraic.

For $a, b \in K_n$ with a compact and b compact-below $f(a)$, let $[a \Rightarrow b]$ denote the usual compact step map which sends arguments above a to b and arguments below a to bottom. Finite joins of such step maps are still compact. Let $\mathcal{A}_f$ be the set of all finite joins of compact step maps lying below f. Then $\mathcal{A}_f$ is directed under inclusion of the underlying finite families, and every member of $\mathcal{A}_f$ is compact-below f.

To see that f is the least upper bound of $\mathcal{A}_f$, fix $x \in K_n$. Since K_n is algebraic, the value $f(x)$ is the least upper bound of the compact elements below it. Each such compact element is already captured by some step approximation built from a compact approximation to x. Hence evaluation at x of the directed family $\mathcal{A}_f$ is cofinal in the compact-below approximants to $f(x)$, so its pointwise supremum is exactly $f(x)$. Since this holds for every x, $f = \sup \mathcal{A}_f$, and K_{n+1} is algebraic.

Thus every finite stage K_n is an algebraic bounded-complete domain.

Passage to the inverse limit. For $x \in K_\infty$, define its nth finite-stage approximation by

$$a_n(x) := f_{n,\infty}(\pi_n(x)),$$

where $f_{n,\infty} : K_n \to K_\infty$ is the canonical exact embedding of the nth stage into the limit. By construction,

$$\pi_m(a_n(x)) = \pi_m(x) \qquad (m \leq n),$$

so each $a_n(x)$ agrees with x on the first $n+1$ coordinates; in particular $a_n(x) \leq x$. Moreover the family $(a_n(x))_{n\geq 0}$ is directed, since passing from stage n to stage $n+1$ only refines the approximation.

Now let $c \ll x$ be compact-below in K_∞. By Lemma 7.7, there is some stage n and some compact $b \ll \pi_n(x)$ in K_n such that

$$c \leq f_{n,\infty}(b) \leq a_n(x).$$

Hence the directed family $\{a_n(x)\}_{n\geq 0}$ is cofinal among compact approximants below x. Therefore x is a least upper bound of this family, and the chosen supremum of the family is equivalent to x in the induced homotopy order:

$$x \simeq \sup_{n\geq 0} a_n(x).$$

Thus every element of K_∞ is a least upper bound of compact elements below it. This proves algebraicity of K_∞.

Finally, bounded completeness of K_∞ is obtained coordinatewise. If $X \subseteq K_\infty$ is upper-bounded, then each projected family $\pi_n(X) \subseteq K_n$ is upper-bounded and therefore has a least upper bound in K_n. Lemma 7.8 shows that these coordinatewise suprema again form a coherent thread, which is the least upper bound of X in K_∞. □

7.2. Exact reflexive packaging

The central structural fact is that the inverse limit is itself reflexive. In fact, the projection-pair machinery on the tower produces global reify and reflect maps. The proof of this fact assembles three technical ingredients that are implicit in the inverse-limit construction but are worth isolating as standalone statements, both because they clarify the argument and because they are reused in the proof-relevant witness analysis of Section 8.

Lemma 7.10 (Density of finite-stage approximants) For every $x \in K_\infty$, define the *nth finite-stage approximant*

$$a_n(x) := f_{n,\infty}(\pi_n(x)) \in K_\infty.$$

Then:

(i) $a_n(x) \leq x$ for every n,
(ii) the family $(a_n(x))_{n\geq 0}$ is directed, and
(iii) $x \simeq \sup_{n\geq 0} a_n(x)$.

In particular, K_∞ is generated, up to chosen-supremum equivalence, by the embedded finite stages $\bigcup_n f_{n,\infty}(K_n)$.

Proof Clause (i) holds because $a_n(x)$ agrees with x on coordinates 0 through n (by the retract property $\pi_n \circ f_{n,\infty} = \mathrm{id}$) and is $\leq x$ on all higher coordinates (by iterated application of the section inequality $f_m^+ f_m^- \leq \mathrm{id}$). Clause (ii) follows because passing from stage n to $n+1$ only refines the approximation: $a_n(x) \leq a_{n+1}(x)$ by the compatibility of the projection pairs. Clause (iii) was established in the proof of Proposition 4.1 (algebraicity of K_∞): every compact element below x is already below some $a_n(x)$, so the directed family is cofinal among the compact approximants to x, hence has x as a least upper bound; the chosen supremum is therefore equivalent to x. □

Lemma 7.11 (Coherence of stage restrictions) Let $g : K_\infty \to K_\infty$ be a continuous endomap and define its *stage-n restriction* by

$$r_n(g) := \pi_n \circ g \circ f_{n,\infty} \ \in \ [K_n \to K_n] = K_{n+1}.$$

Then the family $(r_n(g))_{n\geq 0}$ satisfies the coherence condition

$$f_n^-\big(r_{n+1}(g)\big) = r_n(g) \qquad \text{for all } n \geq 0.$$

In particular, the shifted thread $\big(f_0^-(r_0(g)),\, r_0(g),\, r_1(g),\, \ldots\big)$ is a well-defined element of K_∞.

Proof Fix $a \in K_n$. Applying f_n^- to the map $r_{n+1}(g)$ and evaluating at a gives

$$f_n^-\big(r_{n+1}(g)\big)(a) = \pi_n\big(g(f_{n+1,\infty}(f_n^+(a)))\big).$$

Since $f_{n+1,\infty} \circ f_n^+$ and $f_{n,\infty}$ produce the same coherent thread in K_∞ (both agree on all coordinates up to n and satisfy the tower coherence above n), continuity of g gives $g(f_{n+1,\infty}(f_n^+(a))) = g(f_{n,\infty}(a))$. Projecting to level n yields $r_n(g)(a)$. Since a was arbitrary, $f_n^-(r_{n+1}(g)) = r_n(g)$ as required. The "in particular" clause follows because this coherence condition is exactly the defining property of a thread in the inverse limit K_∞. □

Lemma 7.12 (Monotonicity of application shadows) For $x, y \in K_\infty$, define

$$k_n(x,y) := f_{n,\infty}\big(\pi_{n+1}(x)(\pi_n(y))\big) \in K_\infty.$$

Then

$$k_n(x,y) \leq k_{n+1}(x,y) \qquad \text{for all } n \geq 0.$$

Consequently the directed supremum

$$k(x,y) := \sup_{n\geq 0} k_n(x,y)$$

exists in K_∞.

Proof By thread coherence of x, the coordinate $\pi_n(x)$ is the f_n^--projection of $\pi_{n+1}(x)$. Applying the section inequality $f_n^+ f_n^- \leq \mathrm{id}$ at stage $n+1$ shows that evaluation at the lifted argument in stage $n+1$ dominates the stage-n value. Embedding those values back into K_∞ via the compatible maps $f_{n,\infty}$ and $f_{n+1,\infty}$ yields $k_n(x,y) \leq k_{n+1}(x,y)$. □

Lemma 7.13 (Agreement on embedded finite stages) Let $f, g : K_\infty \to K_\infty$ be continuous endomaps. Suppose that for every $n \geq 0$ and every $y \in K_n$ one has

$$f(f_{n,\infty}(y)) = g(f_{n,\infty}(y)).$$

Then $f = g$.

Proof Fix $z \in K_\infty$. By Lemma 7.10, the finite-stage approximants $a_n(z) = f_{n,\infty}(\pi_n(z))$ form a directed family whose chosen supremum is equivalent to z. By hypothesis, f and g agree on every $a_n(z)$. Since both maps are continuous, they agree on the chosen supremum of that family. Thread equality in K_∞ is coordinatewise (Remark 7.5), and antisymmetry at each finite stage turns the chosen-supremum equivalence back into literal equality with z. Hence $f(z) = g(z)$. Since z was arbitrary, $f = g$. □

Lemma 7.14 (Coordinatewise continuity into the inverse limit) Let A be a complete homotopy partial order and let $g_n : A \to K_n$ be continuous maps satisfying

$$f_n^-(g_{n+1}(a)) = g_n(a) \qquad \text{for every } a \in A \text{ and every } n \geq 0.$$

Then the map

$$a \longmapsto (g_n(a))_{n \geq 0}$$

defines a continuous function $A \to K_\infty$.

Proof The tuple $a \mapsto (g_n(a))_n$ is continuous coordinatewise by hypothesis. The displayed coherence equations show that its image lands in the inverse limit. Since the order and topology on K_∞ are coordinatewise, this coherent tuple defines a continuous map into K_∞. □

The following theorem is therefore reduced to five explicit ingredients: Lemma 7.10 for density of finite-stage approximants, Lemma 7.11 for well-definedness of the reify map, Lemma 7.12 for the directed application shadow, and Lemma 7.13 for equality of continuous maps from dense finite-stage agreement, together with Lemma 7.14 for continuity of coherent coordinate families into the inverse limit.

Theorem 7.15 (Exact K_∞ reflexive packaging) There exist globally continuous maps

$$h : [K_\infty \to K_\infty] \to K_\infty, \qquad k : K_\infty \to [K_\infty \to K_\infty]$$

such that

$$k \circ h = \mathrm{id}_{[K_\infty \to K_\infty]}, \qquad h \circ k = \mathrm{id}_{K_\infty}.$$

The induced application operation

$$\mathsf{app}(x, y) := k(x)(y)$$

is jointly continuous. If $f_{n,\infty} : K_n \to K_\infty$ denotes the canonical stage embedding, then for every $x \in K_\infty$, every $y \in K_n$, and every $n \geq 0$,

$$\pi_n\big(\mathsf{app}(x, f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

Proof We construct h and k directly from the finite stages.

Step 1: exact finite-stage embeddings. Definition 7.3 and Lemma 7.4 provide the canonical embeddings

$$f_{n,\infty} : K_n \to K_\infty$$

together with the exact identities

$$\pi_n \circ f_{n,\infty} = \mathrm{id}_{K_n}, \qquad f_{n+1,\infty} \circ f_n^+ = f_{n,\infty}.$$

Step 2: finite-stage restrictions of endomaps. Given a continuous endomap $g : K_\infty \to K_\infty$, define its stage-$n$ restriction by

$$r_n(g) := \pi_n \circ g \circ f_{n,\infty} \in [K_n \to K_n] = K_{n+1}.$$

These restrictions are compatible:

$$f_n^-\big(r_{n+1}(g)\big) = r_n(g).$$

Indeed, for $a \in K_n$,

$$f_n^-\big(r_{n+1}(g)\big)(a) = \pi_n\big(g(f_{n+1,\infty}(f_n^+(a)))\big) = \pi_n\big(g(f_{n,\infty}(a))\big) = r_n(g)(a),$$

because $f_{n+1,\infty} \circ f_n^+$ and $f_{n,\infty}$ define the same thread in the inverse limit.

The coherence condition of Lemma 7.11 guarantees that we may assemble the coherent thread

$$h(g)_0 := f_0^-(r_0(g)), \qquad h(g)_{n+1} := r_n(g) \quad (n \geq 0).$$

Each coordinate $g \mapsto h(g)_n$ is continuous because it is built by composing continuous maps, and Lemma 7.11 supplies the required thread equations. Lemma 7.14 therefore yields a continuous map

$$h : [K_\infty \to K_\infty] \to K_\infty.$$

Step 3: finite-stage application shadows. For $x, y \in K_\infty$, define the nth binary approximation to application by

$$k_n(x, y) := f_{n,\infty}\big(\pi_{n+1}(x)(\pi_n(y))\big) \in K_\infty.$$

Each k_n is jointly continuous, and Lemma 7.12 shows that the family $(k_n)_{n\geq 0}$ is monotone. Hence the directed supremum

$$k(x, y) := \sup_{n\geq 0} k_n(x, y)$$

exists in K_∞. For each fixed coordinate m, the map

$$(x, y) \longmapsto \pi_m(k(x, y))$$

is the directed supremum of the continuous coordinate maps $(x, y) \mapsto \pi_m(k_n(x, y))$, hence is continuous. The stage-embedding identities of Lemma 7.4 and the monotonicity from Lemma 7.12 ensure that these coordinates satisfy the inverse-limit coherence equations. Lemma 7.14 therefore gives a jointly continuous map

$$(x, y) \longmapsto k(x, y) \in K_\infty,$$

and currying gives

$$k : K_\infty \to [K_\infty \to K_\infty].$$

Step 4: the stagewise formula. Fix $x \in K_\infty$ and $y \in K_n$. Restrict the map $k(x)$ along the exact embedding $f_{n,\infty}$. On the one hand, the nth approximant $k_n(x, f_{n,\infty}(y))$ has nth coordinate exactly $\pi_{n+1}(x)(y)$. On the other hand, every higher approximant has nth coordinate below the same value, so after projecting to level n the entire directed family is bounded above by $\pi_{n+1}(x)(y)$ and contains an element attaining that upper bound. Since projection preserves directed suprema,

$$\pi_n\big(k(x)(f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

This is exactly the stated stagewise application formula.

Step 5: the section law $k \circ h = \mathrm{id}$. Let $g : K_\infty \to K_\infty$. For every stage n and every $y \in K_n$, the previous formula applied to $h(g)$ gives

$$\pi_n\big(k(h(g))(f_{n,\infty}(y))\big) = \pi_{n+1}(h(g))(y) = r_n(g)(y) = \pi_n\big(g(f_{n,\infty}(y))\big).$$

Thus $k(h(g))$ and g agree on every embedded finite stage, and Lemma 7.13 implies

$$k(h(g)) = g.$$

Step 6: the retract law $h \circ k = \mathrm{id}_{K_\infty}$. For $x \in K_\infty$ and $n \geq 0$,

$$\pi_{n+1}(h(k(x))) = r_n(k(x)).$$

Evaluating this map at $y \in K_n$ and using the stagewise formula just proved, we obtain

$$\pi_{n+1}(h(k(x)))(y) = \pi_n\big(k(x)(f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y).$$

Therefore the $(n + 1)$st coordinates of $h(k(x))$ and x coincide for all $n \geq 0$. At level 0 one gets

$$\pi_0(h(k(x))) = f_0^-(\pi_1(x)) = \pi_0(x)$$

by thread coherence. Hence, by Remark 7.5,

$$h(k(x)) = x.$$

Finally, $\mathsf{app}(x, y) = k(x)(y)$ is jointly continuous because it is the composite of the continuous map

$$(x, y) \longmapsto (k(x), y)$$

with the continuous evaluation map on the function space $[K_\infty \to K_\infty]$. $\square$

Remark 7.16 (Relation to classical projection-pair models) Classical projection-pair inverse-limit constructions already motivate reflexive objects such as D_∞ [13, 1]. The two-sided isomorphism $h \circ k = \mathrm{id}$ and $k \circ h = \mathrm{id}$ is therefore not claimed here as a new abstract inverse-limit theorem. The specific new content of Theorem 7.15 is the stagewise evaluation formula

$$\pi_n\big(\mathsf{app}(x, f_{n,\infty}(y))\big) = \pi_{n+1}(x)(y),$$

which characterizes application at every finite stage and is used in Section 8 to pin down the witness interpretations at the level of K_0 coordinates. The explicit formulas for h and k given in the proof also provide a constructive, self-contained verification of this concrete K_∞ instance rather than an appeal to the general inverse-limit theorem alone.

The maps h and k are implemented in the Lean formalization by projection-pair data at each finite stage. The following code fragment illustrates that interface; Section 8 then uses the stagewise formula of Theorem 7.15 to separate proof-relevant witness classes inside K_∞.

Listing 1. Projection pairs for the K_n tower (from `KInfinity/Construction.lean`).

```
structure ProjectionPair
    (A : CompleteHomotopyPartialOrder)
    (B : CompleteHomotopyPartialOrder) where
  emb : ContinuousMap A B
  proj : ContinuousMap B A
  retract : ContinuousMap.comp proj emb = ContinuousMap.id A
  section_le :
    forall x : B.Obj, B.Rel (emb (proj x)) x
```

8. Fixed-Span Witness Classification and Separation

Paper II [10] established, via the witness-separation span restated below as Example 8.1, that the K_∞ model is nontrivial: a distinguished β/η witness pair from the same redex receives semantically distinct images. What it did not do was analyze the fixed forward witness language on that span or trace, in the canonical higher tower on K_∞, how the resulting point-level separation propagates through all higher dimensions. The present section remains deliberately local: it is a theorem about this fixed span, not a global witness calculus for arbitrary source/target pairs.

The present section has two logically distinct parts. First, we introduce a *witness language* that remembers which contraction was chosen and, on the fixed witness-separation span, classifies every witness in the chosen forward fragment by a canonical β or η tag. This is the model-specific input. Second, we show that once the resulting point-level dichotomy is established, the absence of connecting 1-cells and higher cells is forced by the equality-generated structure of the K_∞ higher-cell tower, including the recursively completed dimensions.

8.1. A basic witness-separation example

Paper II [10] proved the non-triviality of K_∞ by means of the following concrete span, which we restate here for direct reference.

Example 8.1 (Witness-separation span) Consider the source

$$M = ((\lambda z.\, xz)\, y)$$

and the common target

$$N = xy.$$

The two basic one-step contractions are displayed explicitly as

$$((\lambda z.\, xz)\, y) \;\to_\beta\; xz[y/z] = xy$$

and

$$((\lambda z.\, xz)\, y) \;\to_\eta\; xy,$$

where in the second line the η-contraction is performed in the function part $\lambda z.\, xz \to_\eta x$. Thus the same source and target admit two distinguished one-step witnesses: the direct β-contraction and the direct η-contraction through the function part.

Remark 8.2 (Notation for the running span) Although the formal development works uniformly with de Bruijn indices, we display Example 8.1 in named-variable notation for readability. The free variables x and y should therefore

be read only as mnemonic placeholders for a fixed open term; all subsequent classification and semantic arguments remain statements about the de Bruijn formalism fixed in Section 2.

The point of the proof-relevant refinement is that we do not merely remember that the endpoints are propositionally $\beta\eta$-equal; we remember *which contraction witness was chosen.* A general proof-relevant explicit 1-witness from M to N may contain reflexive padding before or after the essential contraction, but within this witness-separation span the witness language still falls into exactly two reachability classes: the β-class and the η-class. Thus every proof-relevant 1-witness on this span carries a canonical tag

$$\mathrm{tag}(t) \in \{\beta, \eta\}.$$

Figure 2 records this fixed span and the two canonical semantic images attached to its witness classes.

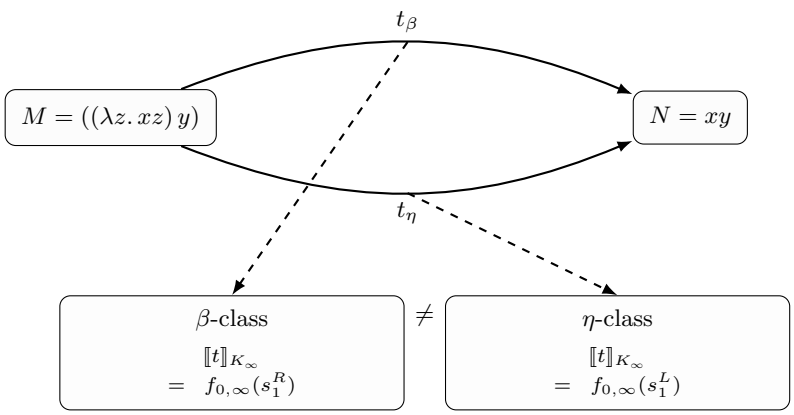


Figure 2 The witness-separation span of Example 8.1. The same source and target admit a β-witness and an η-witness, but their canonical K_∞ interpretations land at distinct points.

8.2. Witness languages

Definition 8.3 (Fixed-span 1-term witnesses) Fix the source M and target N of Example 8.1. Let $W(A, B)$, for $A, B \in \{M, N\}$, be the inductively generated typed family of witnesses generated by:

(1)the direct witnesses $t_\beta : W(M, N)$ and $t_\eta : W(M, N)$;
(2)the reflexive witnesses $\mathsf{refl}_M : W(M, M)$ and $\mathsf{refl}_N : W(N, N)$;
(3)composition: if $u : W(A, B)$ and $v : W(B, C)$, then $u \cdot v : W(A, C)$.

A 1-*term witness* from M to N is an element of $W(M, N)$. There is no inverse constructor in this language.

Remark 8.4 (Scope of the witness language) The language $W(A, B)$ is intentionally much smaller than the full reduction-sequence language or the full low-dimensional higher-conversion syntax. It keeps only the two forward one-step witnesses on the fixed span, reflexivity, and composition. In particular, it does not attempt to classify arbitrary reduction sequences, inverses, whiskering, or congruence data. Theorem 8.7 is therefore a fixed-span statement about this forward fragment only.

In this section we restrict attention to the fixed witness-separation span of Example 8.1

$$M = ((\lambda z.\, xz)\, y), \qquad N = xy.$$

To each explicit 1-term witness t from M to N we attach two pieces of semantic data. This interpretation is intentionally local: it classifies the fixed-span witness language $W(M, N)$ by the two distinguished points of K_∞ attached to Example 8.1.

(1) a continuous equality ϵ_t witnessing that the source and target terms have the same ordinary denotation in the reflexive semantics, and
(2) a distinguished point of K_∞ recording whether the witness belongs to the β-class or to the η-class.

The second piece is what detects proof relevance. Let s_1^R and s_1^L be the right and left poles in the S^1-summand of the non-trivial base object used to define $K_0 = N^+$. If $\mathrm{tag}(t) = \beta$, we set

$$[\![t]\!]_{K_\infty} := f_{0,\infty}(s_1^R),$$

and if $\mathrm{tag}(t) = \eta$, we set

$$[\![t]\!]_{K_\infty} := f_{0,\infty}(s_1^L).$$

Thus the interpretation depends only on the essential witness class, not on the amount of reflexive padding surrounding it. We now make this classification and its stability precise before stating the main separation theorem.

Lemma 8.5 (Witness classification on the witness-separation span) Let t be an explicit 1-term witness from the source $M = ((\lambda z.\, xz)\, y)$ to the target $N = xy$. Then exactly one of the following holds:

(a) t is obtained from the direct β-contraction by inserting reflexive witnesses on the left and/or right;
(b) t is obtained from the direct η-contraction by inserting reflexive witnesses on the left and/or right.

In particular, the canonical tag $\mathrm{tag}(t) \in \{\beta, \eta\}$ is well-defined.

Proof Proceed by structural induction on the explicit witness language. The base constructors are the direct β- and η-witnesses together with reflexivity. The reflexive case cannot occur between the chosen source and target because $M \neq N$, so any witness from M to N must be classified by one of the two nontrivial contractions.

For a composite witness, once a nontrivial β- or η-witness has appeared, the remaining factor can contribute only reflexive padding. Indeed, the witness-separation span admits no third nontrivial contraction pattern: the source has exactly the direct outer β-contraction and the inner η-contraction through the function part, and the target xy supports no further forward β- or η-step. Hence every sequential composite factors as one distinguished nontrivial contraction together with reflexive witnesses on the left and/or right.

The two classes are disjoint because the direct β- and η-steps are syntactically distinct and have different chosen semantic tags. Therefore exactly one of (a) or (b) holds, and the tag is well-defined. □

Corollary 8.6 (Stability under reflexive padding) Let t and t' be explicit 1-term witnesses from M to N that differ only by insertion or removal of reflexive subpaths. Then $\mathrm{tag}(t) = \mathrm{tag}(t')$, and hence $[\![t]\!]_{K_\infty} = [\![t']\!]_{K_\infty}$.

Proof By Lemma 8.5, the two classes are defined precisely by adjoining reflexive witnesses to the direct β- or η-contraction. Inserting or removing reflexive padding therefore cannot change whether a witness is of β-type or η-type. Hence $\mathrm{tag}(t) = \mathrm{tag}(t')$, and the K_∞-interpretation was defined in terms of the tag alone. □

With these preliminaries in hand, we can state the main fixed-span persistence theorem.

Theorem 8.7 (Fixed-span witness classification and separation in the canonical K_∞ tower) Let t_β and t_η be the explicit β_1- and η_1-witnesses from Example 8.1. In the canonical higher tower on K_∞ described in Remark 7.6, their canonical interpretations are:

(i) distinct as K_∞-points,
(ii) not connected by any 1-cell, i.e., by any equality in the carrier of K_∞,
(iii) consequently not connected by any higher cell in any positive dimension of that tower.

The same conclusion holds for any pair of witnesses in the fixed-span language $W(M, N)$ whose canonical tags are β and η, respectively.

Proof By Lemma 8.5, every witness in the fixed-span language $W(M, N)$ lies in exactly one of the two classes obtained from the direct β- or η-contraction by reflexive padding, and Corollary 8.6 shows that the tag is invariant under insertion or removal of those reflexive subpaths. Thus every witness with β-tag is interpreted by the point $f_{0,\infty}(s_1^R)$, while every witness with η-tag is interpreted by $f_{0,\infty}(s_1^L)$.

For the distinguished witnesses t_β and t_η, we therefore have

$$\pi_0([\![t_\beta]\!]_{K_\infty}) = s_1^R, \qquad \pi_0([\![t_\eta]\!]_{K_\infty}) = s_1^L.$$

Since s_1^R and s_1^L are distinct poles of the chosen S^1-summand, the two interpreted points are distinct in K_∞. This proves (i).

By Remark 7.6, a 1-cell in the canonical K_∞ tower is literally an equality in the carrier of K_∞. Hence the distinct points $[\![t_\beta]\!]_{K_\infty}$ and $[\![t_\eta]\!]_{K_\infty}$ admit no 1-cell, proving (ii). Higher cells in that tower are iterated equalities between parallel lower cells, so once the boundary 1-cell type is empty there can be no higher cell in any positive dimension with these endpoints. This proves (iii).

The final sentence follows because every witness in $W(M, N)$ with β-tag is interpreted by the same canonical β-point and every witness with η-tag by the same canonical η-point. □

Remark 8.8 Theorem 8.7 should be read as a fixed-span persistence statement. Its model-specific input is the point-level β/η distinction on the witness language of Example 8.1; the higher-dimensional non-connection is then forced by the canonical identity-type tower chosen on K_∞. In particular, the theorem does not claim a global compositional witness semantics for arbitrary pairs of terms, nor does it classify alternative higher structures that one might place on K_∞.

9. Discussion and Future Work

9.1. Summary of results

We have presented four theorem packages on the theory of higher λ-models:

(A) Structural comparison with the recursive completion for higher λ-conversions (Theorem 5.6).
(B) A smaller sufficient semantic interface for associator/pentagon reasoning (Theorem 6.8).
(C) Fixed-span witness classification and separation in the canonical K_∞ tower (Theorem 8.7).
(D) Exact reflexive packaging and application for K_∞ (Theorem 7.15).

The four theorems do not all have the same rôle. Theorems 6.8 and 7.15 contain the main generic-semantic and concrete-model input. Theorem 5.6 records the finite packaging boundary between the explicit syntax and its recursive continuation, while Theorem 8.7 explains what the fixed-span point-level separation implies inside the canonical higher tower on K_∞. The net claim of the paper is therefore not four co-equal breakthroughs, but a division of labour between a small semantic input and the structural consequences forced by the chosen towers.

9.2. Comparison with related work

We organize the discussion into three threads: direct antecedents within the higher λ-model programme, the broader higher-categorical and homotopical background, and the classical domain-theoretic lineage.

Direct antecedents. The immediate foundation is the two-paper programme of Martínez-Rivillas and de Queiroz. Paper I [11] introduced the extensional Kan semantics, defined the explicit tower of n-conversions, and proved the main inclusion $\mathrm{TH}_{\lambda=} \subseteq$ HoTFT. Paper II [10] constructed K_∞ and already isolated the distinguished β/η witness pair restated here as Example 8.1. The present work begins from those results but addresses four specific remaining gaps. For Theorem 5.6, the missing point is not the existence of recursive higher-derivation syntax, but the comparison between the explicit low-dimensional tower and the equality-generated continuation together with the proof of strict globular compatibility for that continuation. For Theorem 6.8, the missing point is the isolation of the smaller WLWR-plus-front-seed interface sufficient for the later semantic arguments. For Theorem 8.7, Paper II gives only point-level separation; what was missing was a witness-level classification on the witness-separation span together with an account of how that separation behaves in the canonical higher tower on K_∞. For Theorem 7.15, the missing point is the exact reflexive packaging of K_∞ by globally continuous reify, reflect, and application maps with explicit stagewise formulas and exact coordinate identities.

Earlier in the same programme, Martínez-Rivillas and de Queiroz [9] developed a homotopy domain theory that enriches the classical Scott semantics with an explicit homotopical layer, and in [12] they studied the ∞-groupoid generated by an arbitrary topological λ-model. Both papers anticipate the idea that the conversion structure of the λ-calculus has a nontrivial higher-categorical life, but neither constructs the realization comparison of Theorem 5.6 or the fixed-span witness classification of Theorem 8.7.

A closely related syntactic thread is the computational-paths programme of de Veras, Ramos, de Queiroz, and de Oliveira [4], which showed that the paths generated by directed reduction steps in a rewriting system form a weak groupoid. Their construction provides a syntactic template for the 1-dimensional layer of our tower: the reduction sequences between λ-terms are precisely computational paths, and the explicit 2-cells (homotopies between reduction sequences) can be viewed as the coherence data that upgrade the resulting path groupoid to a 2-groupoid. The present paper pushes that picture beyond the path-groupoid layer by comparing the recursive completion of the low-dimensional core with the explicit higher-conversion tower (Theorem 5.6).

Higher-categorical and homotopical background. The broader higher-categorical references are conceptual rather than technical antecedents. Batanin [3] provides the globular language relevant to our use of source/target compatibility, whiskering, and coherence, but we do not import his machinery directly. Likewise, Grothendieck [5], homotopy type theory [6], and Lurie [8] are cited only as background reminders that higher identification data can carry nontrivial coherent structure. The present

paper does not derive results from those frameworks; it studies a concrete syntactic/semantic construction for the untyped λ-calculus. We intend these comparisons as orientation, not as claims of direct technical overlap.

Classical domain-theoretic lineage. The K_∞ construction (Section 7) is a direct descendant of Scott's continuous lattice semantics [13] and the later projection-pair account surveyed by Abramsky and Jung [1]. Scott showed that every countably based continuous lattice admits a reflexive structure, and the resulting models interpret the untyped λ-calculus. The K_∞ tower replaces Scott's single-step reflexivity with an iterated function-space tower $K_{n+1} = [K_n \to K_n]$ whose inverse limit inherits both the classical algebraic-domain structure and a new homotopical layer.

Barendregt's [2] systematic treatment of λ-models provides the classical reference frame: the notions of extensionality, the $\beta\eta$-theory, Church–Rosser, and confluence that appear in our preliminaries and baseline sections are standard in the sense of [2]. The novelty of the present approach is that these classical ingredients are enriched with higher-dimensional semantic structure that was invisible in the purely propositional setting.

Nature of the contribution. The rôle of the formal development is methodological and certifying rather than expository. The mathematical burden of the paper is carried by the arguments in Sections 5–8. Among those, Theorem 7.15 is the strongest model-specific statement, Theorem 6.8 identifies a smaller coherence input currently proved sufficient for the later semantic comparison arguments, and Theorems 5.6 and 8.7 describe structural consequences of the recursive higher tower once the low-dimensional and point-level data are fixed. Lean 4 [7] provided a disciplined setting in which these arguments were discovered and checked, but the paper is intended to stand or fall on its mathematics rather than on the existence of the formalization. In the broader logic-and-computation tradition, the results illustrate a recurring theme: that the proof-relevant refinement of a classical equivalence relation (here, $\beta\eta$-convertibility) reveals computational structure—witness classes, coherence data, semantic separation—that the propositional collapse erases. The constructive character of the proofs, particularly the explicit witness constructions, aligns with the programme of making the computational content of logical derivations visible and tractable. In short: higher λ-models can now be read as a small coherence seed, an exact concrete inverse-limit model, and the recursive higher consequences forced by those data.

9.3. Open problems

1. **Eliminating the front-seed.** Derive the front-seed coherence package (seeds FS1 and FS2 in Theorem 6.8) directly from the bare extensional Kan complex interface. A positive answer would unify the bare and coherent semantic regimes, showing that every extensional Kan complex is automatically coherent in the sense needed for associator and pentagon reasoning.
2. **Alternative higher towers on K_∞.** The present paper uses the canonical identity-type tower on the carrier of K_∞. It would be interesting to compare this choice with more geometric higher structures, for example towers built from path objects or from the Scott topology, and to ask whether the fixed-span separation persists or strengthens there.
3. **Typed extensions.** Extend the recursive-completion comparison and its higher-dimensional consequences to typed λ-calculi, where the type structure could provide additional constraints on the higher-cell tower. Simply-typed and polymorphic calculi are natural first targets.
4. **Comparison questions with HoTT.** Investigate whether the low-dimensional core $\mathscr{G}_\lambda$ or its recursive completion admits a useful comparison map to the fundamental higher-groupoid constructions of homotopy type theory [6].
5. **Explicit strictification.** Determine whether the low-dimensional core $\mathscr{G}_\lambda$ or its recursive completion admits a strictification to a strict ω-category, and if so, what information about the λ-calculus is lost in the process.

9.4. Formal verification note

All results proved in this paper have been formally verified in the Lean 4 theorem prover [7]; the complete development is available at

`https://github.com/Arthur742Ramos/HigherLambdaModel`

From the repository root, `lake build` checks the full development. The file `docs/theorem_index.md` is the exhaustive paper-to-Lean claim matrix, and the project sources contain no local uses of `sorry`, `admit`, or `axiom`. From the mathematical point of view, the formalization plays an evidential role: it certifies that every case split, boundary compatibility check, and recursive induction invoked above has been verified mechanically. In particular, the recursive inductions in the proofs of Theorems 5.6 and 7.15 involve several dozen case-by-case verifications that are straightforward but tedious to reproduce on paper; the formal proof provides independent assurance that none were dropped. From a logic-and-computation perspective, the formalization also serves as a computational certificate: the Lean kernel's type-checking algorithm independently verifies every logical step, providing a degree of assurance that complements traditional peer review. The formalization also covers all four main theorems, the baseline results of Section 3, and the shared low-dimensional core through dimension 6 together with the recursively completed tower used in Theorem 5.6.

To keep the present paper focused on the mathematics, we record only one short code fragment in the main text—the `ProjectionPair` interface from Section 7. For readers who want a direct bridge back to the formal development, Appendix A records representative Lean entry points for the four main theorems.

A. Lean roadmap for the main theorems

This appendix gives a compact map from the four headline results of the paper to representative declarations in the Lean development, together with short illustrative excerpts. It is not an exhaustive index; the full paper-to-Lean claim matrix is maintained in `docs/theorem_index.md`. The excerpts below are intentionally short and omit some argument lists with `...` when the omitted data are not mathematically central to the discussion in the paper.

Theorem A: comparison with the recursive completion

Primary files.

```
HigherLambdaModel/Lambda/Coherence.lean
HigherLambdaModel/Lambda/TruncationCore.lean
HigherLambdaModel/Lambda/Truncation.lean
```

Representative declarations.

```
AllDimensionalHigherConversionCoherence
lambdaOmegaConstructiveRealize
lambdaConstructiveHigherConversionCoherence
```

Listing 2. Representative Lean excerpt for Theorem A (from `Lambda/TruncationCore.lean`).

```
def lambdaConstructiveHigherConversionCoherence :
    HigherLambdaModel.Lambda.Coherence.AllDimensionalHigherConversionCoherence
      HigherLambdaModel.Lambda.NTerms.lambdaTower
      lambdaOmegaReflexiveTower where
  realize := lambdaOmegaConstructiveRealize
  source_comm := by
    intro n x
    exact lambdaOmegaConstructiveRealize_source_comm x
  target_comm := by
    intro n x
    exact lambdaOmegaConstructiveRealize_target_comm x
```

Theorem B: front-seed sufficiency

Primary file.

```
HigherLambdaModel/Lambda/ExtensionalKanHigher.lean
```

Representative declarations.

```
FrontSeedCoherentExtensionalKanComplex
FrontSeedCoherentExtensionalKanComplex.reductionSeq_comp_associator_in_Theory3
FrontSeedCoherentExtensionalKanComplex.reductionSeq_pentagon_in_Theory3
FrontSeedCoherentExtensionalKanComplex.homotopy2_pentagon_source_bridge_in_Theory3
FrontSeedCoherentExtensionalKanComplex.homotopy2_pentagon_target_bridge_in_Theory3
FrontSeedCoherentExtensionalKanComplex.homotopy2_pentagon_shell_bridge_in_Theory3
```

The source bridge lands in the structural semantic source shell. The target and shell bridges land in the mixed target shell where the right-whiskered factor remains equality-generated; the fully structural pentagon comparison itself is recorded separately by `FrontSeedCoherentExtensionalKanComplex.reductionSeq_pentagon_in_Theory3`.

Listing 3. Representative Lean excerpt for Theorem B (from `Lambda/ExtensionalKanHigher.lean`).

```
structure FrontSeedCoherentExtensionalKanComplex
    extends ExtensionalKanComplex where
  pentagonInnerRightFrontReflPath3 : ...
```

```
  wlwrPath3 : ...

noncomputable def
    FrontSeedCoherentExtensionalKanComplex.reductionSeq_comp_associator_in_Theory3
    (K : FrontSeedCoherentExtensionalKanComplex) ... :=
  reductionSeq_comp_associator_in_Theory3_ofPentagonInnerRightFrontRefl
    K.toExtensionalKanComplex
    K.pentagonInnerRightFrontReflPath3
    K.wlwrPath3 ...
```

Theorem C: fixed-span persistence in the canonical K_∞ tower

Primary files.

```
HigherLambdaModel/KInfinity/Examples.lean
HigherLambdaModel/KInfinity/Properties.lean
HigherLambdaModel/KInfinity/ContinuousSemantics.lean
HigherLambdaModel/Lambda/NTerms.lean
```

Representative declarations.

```
example42NTerm1WitnessTag
example42NTerm1Witness_interpretation
betaEtaPaper_beta1Witness_interpretation
betaEtaPaper_eta1Witness_interpretation
betaEtaPaper_nterm1Witness_interpretations_distinct
betaEtaPaper_nterm1Witness_interpretations_no_path
betaEtaPaper_nterm1Witness_interpretations_no_2cell
betaEtaPaper_nterm1Witness_interpretations_no_3cell
betaEtaPaper_nterm1Witness_interpretations_no_4cell
betaEtaPaper_nterm1Witness_interpretations_no_5cell
betaEtaPaper_nterm1Witness_interpretations_no_recursive_higher_cell
```

Listing 4. Representative Lean excerpt for Theorem C (from `KInfinity/Examples.lean`).

```
noncomputable def betaEtaPaper_beta1Witness_interpretation :
    Example42NTerm1WitnessInterpretation :=
  example42NTerm1Witness_interpretation betaEtaPaper_beta1Witness

noncomputable def betaEtaPaper_eta1Witness_interpretation :
    Example42NTerm1WitnessInterpretation :=
  example42NTerm1Witness_interpretation betaEtaPaper_eta1Witness

theorem betaEtaPaper_nterm1Witness_interpretations_no_recursive_higher_cell
    (n : Nat) :
    not (Nonempty {x : kInfinityTower.Cell (n + 5) // ... }) := ...
```

Theorem D: exact reflexive packaging of K_∞

Primary file.

```
HigherLambdaModel/KInfinity/Properties.lean
```

Representative declarations.

```
hInfinity
kInfinityContinuous
kInfinityReflexiveCHPO
hInfinity_kInfinityContinuous_apply
kInfinityContinuous_hInfinityContinuous_apply
remark_4_3
```

```
Proposition43Witness
applicationContinuous
```

Listing 5. Representative Lean excerpt for Theorem D (from `KInfinity/Properties.lean`).

```
noncomputable def kInfinityReflexiveCHPO : ReflexiveCHPO KInfinityCHPO where
  reify := hInfinityContinuous
  reflect := kInfinityContinuous
  retract := by
    ext x
    exact hInfinity_kInfinityContinuous_apply x

theorem kInfinityContinuous_hInfinityContinuous_apply
    (f : ContinuousMap KInfinityCHPO KInfinityCHPO) :
    kInfinityContinuous (hInfinityContinuous f) = f := by
  ext x
  exact kInfinityContinuous_hInfinityContinuous_restrict_apply f x

noncomputable def applicationContinuous :
    ContinuousMap (Product.chpo KInfinityCHPO KInfinityCHPO) KInfinityCHPO := ...
```

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